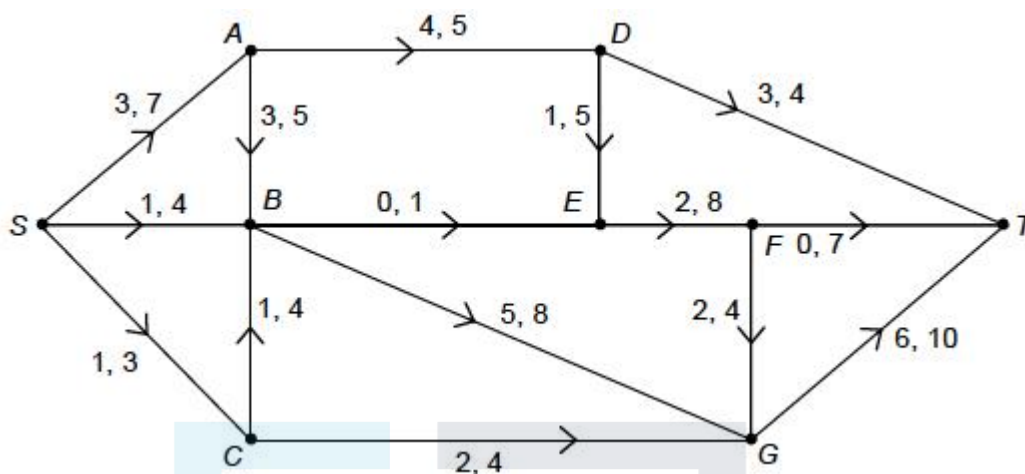


Q1.

The network shows a system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each arc.



- (a) There is a feasible flow from S to T .
- (i) Explain why arc AD must be at its lower capacity. (1)
- (ii) Explain why arc BE must be at its upper capacity. (1)
- (b) Explain why a flow of 11 litres per second through the network is impossible. (1)
- (c) The network in **Figure 1** shows a second system of pipes, where S is the source and T is the sink.

The lower and upper capacities, in litres per second, of each pipe are shown on each edge.

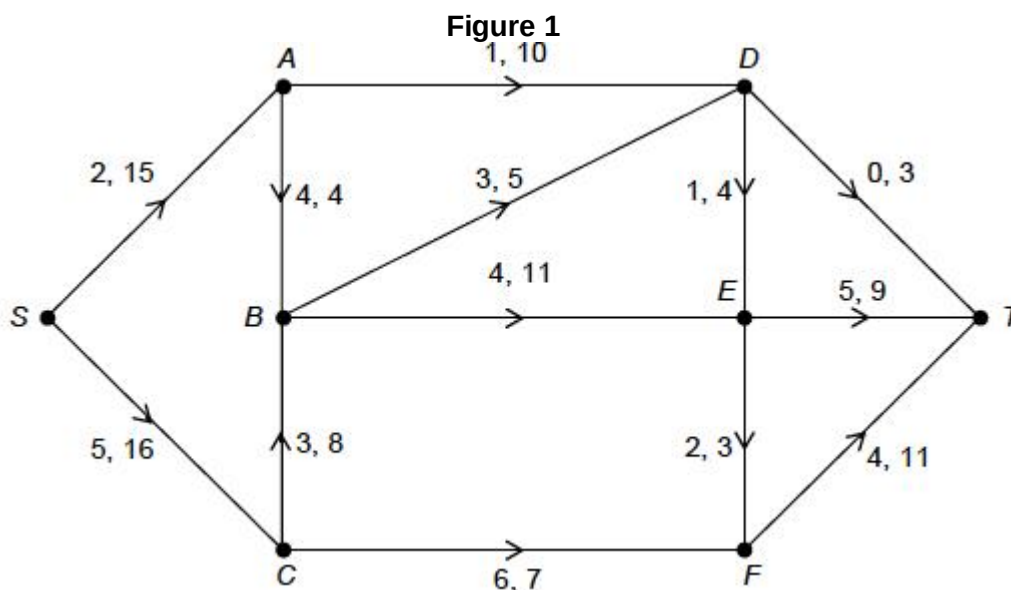
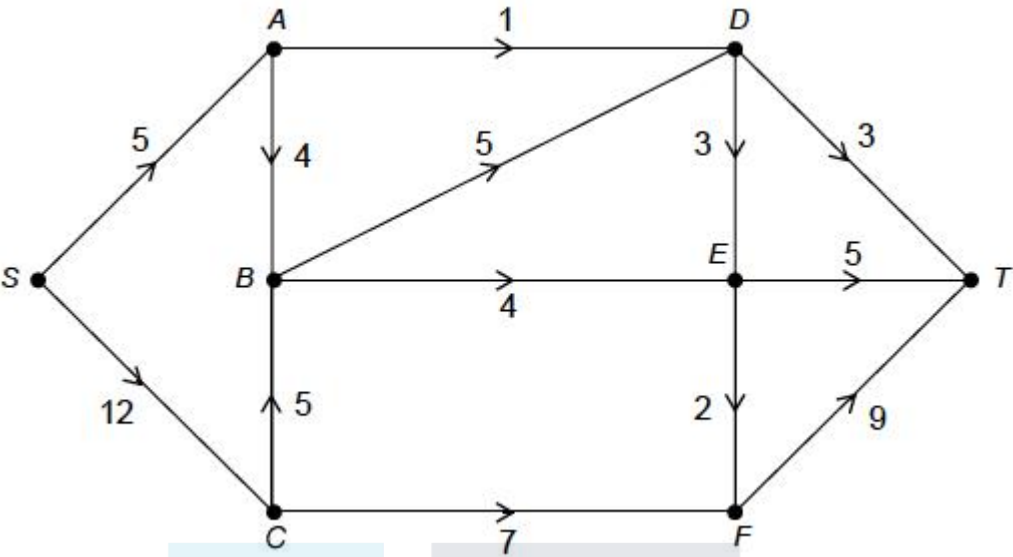


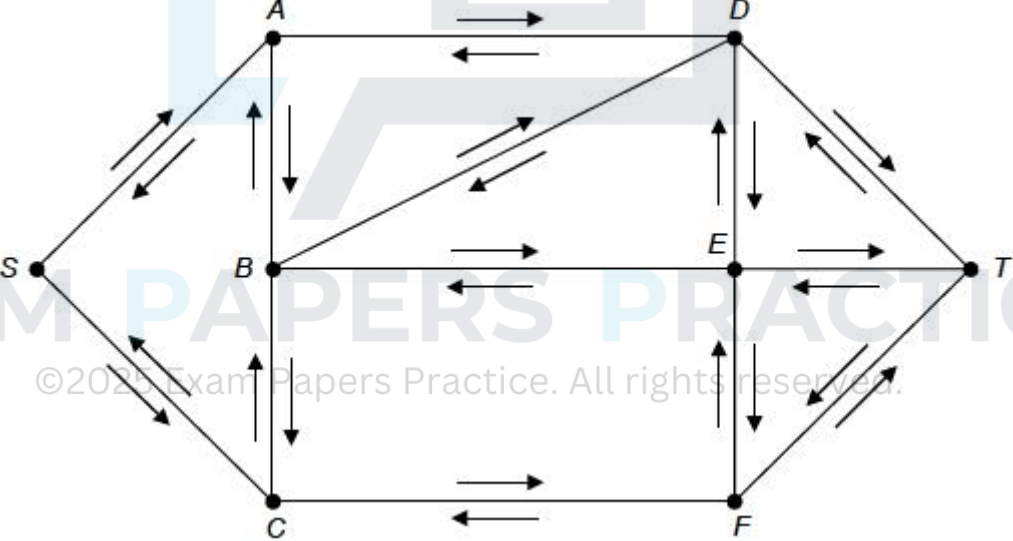
Figure 2 shows a feasible flow of 17 litres per second through the system of pipes.

Figure 2



- (i) Using **Figures 1** and **2**, indicate on **Figure 3** potential increases and decreases in the flow along each arc.

Figure 3



(2)

- (ii) Use flow augmentation on **Figure 3** to find the maximum flow from **S** to **T**.

You should indicate any flow augmenting paths clearly in the table below and modify the potential increases and decreases of the flow on **Figure 3**.

(1)

Augmenting Path	Flow

(iii) Prove the flow found in part (c)(ii) is maximum. (1)

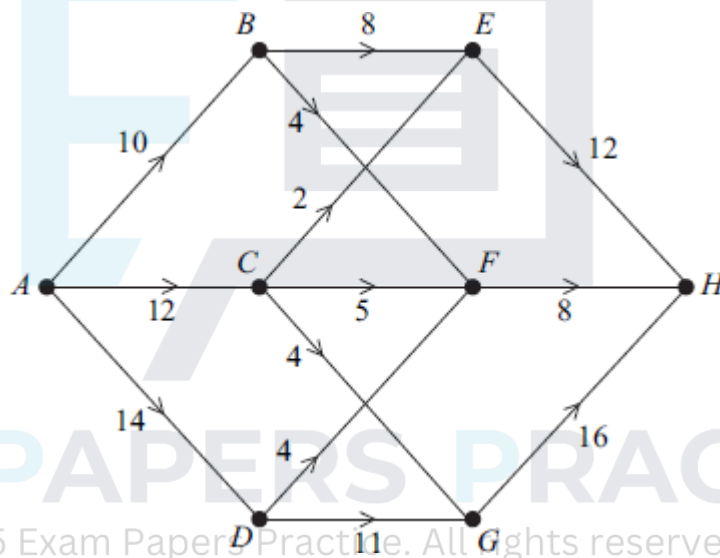
(iv) Due to maintenance work, the flow through node *E* is restricted to 9 litres per second.

Interpret the impact of this restriction on the maximum flow through the system of pipes.

(2)
(Total 11 marks)

Q2.

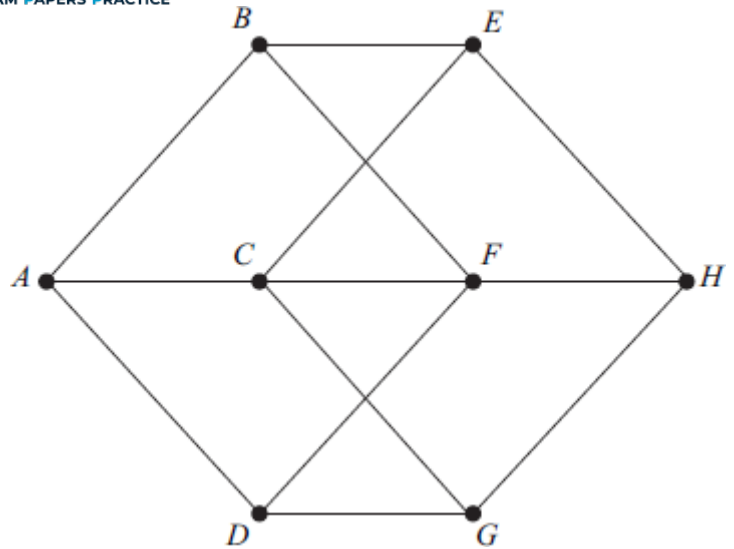
The network below represents a system of pipes. The capacity of each pipe, in litres per second, is indicated on the corresponding edge.



(a) Find the maximum flow along each of the routes *ABEH*, *ACFH* and *ADGH* and enter their values in the table on **Figure 1**.

Figure 1

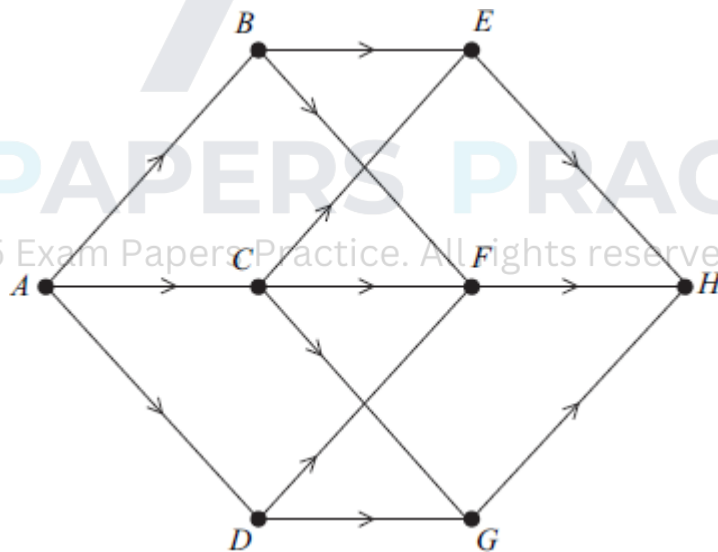
Route	Flow
<i>ABEH</i>	
<i>ACFH</i>	
<i>ADGH</i>	



(1)

- (b) (i) Taking your answers to part (a) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow-augmenting routes in the table and modify the potential increases and decreases of the flow on the network.
- (5)
- (ii) State the value of the maximum flow and, on **Figure 2** below, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



(2)

- (c) Confirm that you have a maximum flow by finding a cut of the same value. List the edges of your cut.

(1)

(Total 9 marks)

Q3.

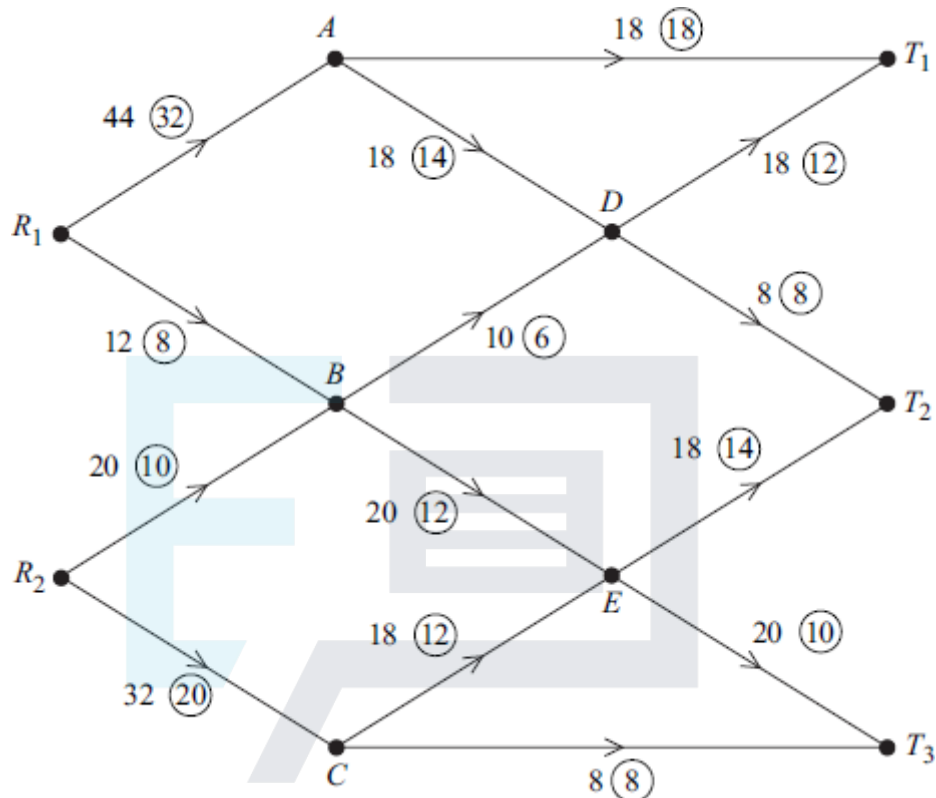
Figure 1 shows a network of pipes.

Water from two reservoirs, R_1 and R_2 , is used to supply three towns, T_1 , T_2 and T_3 .

In **Figure 1**, the capacity of each pipe is given by the number **not** circled on each edge. The numbers in circles represent an initial flow.

- (a) Add a supersource, supersink and appropriate weighted edges to **Figure 1**.

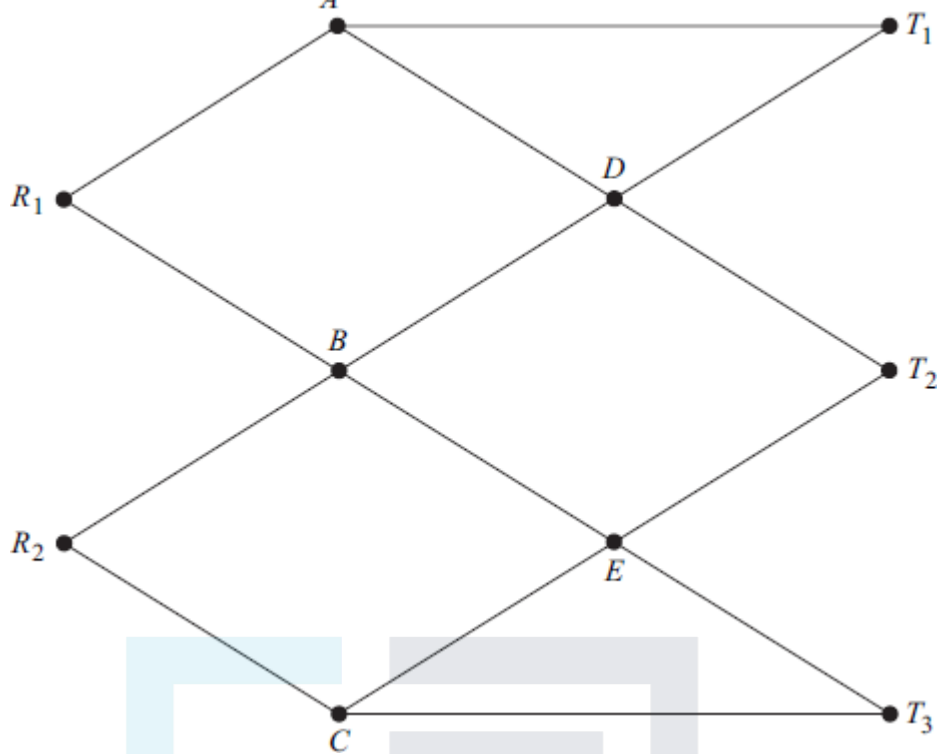
Figure 1



- (b) (i) Use the initial flow and the labelling procedure on **Figure 2** to find the maximum flow through the network.

You should indicate any flow augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

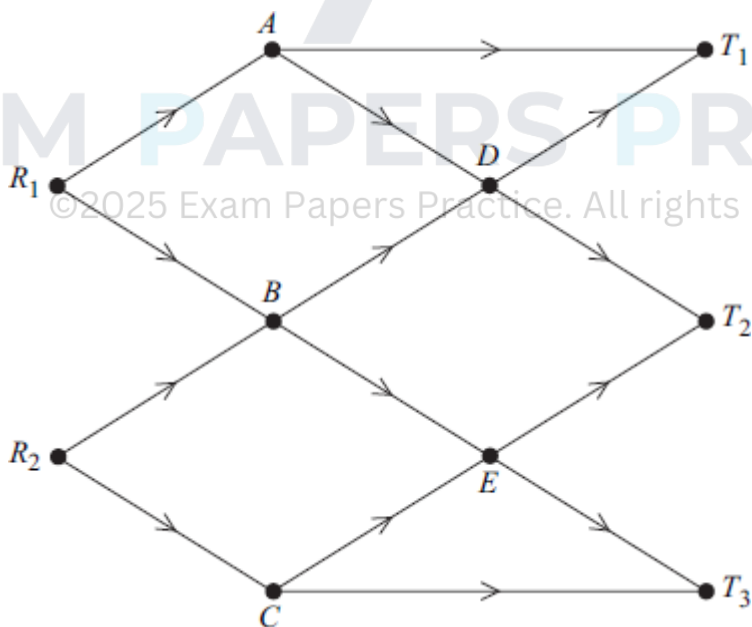
Figure 2



(5)

- (ii) State the value of the maximum flow and, on **Figure 3**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 3



Route	Flow

(2)

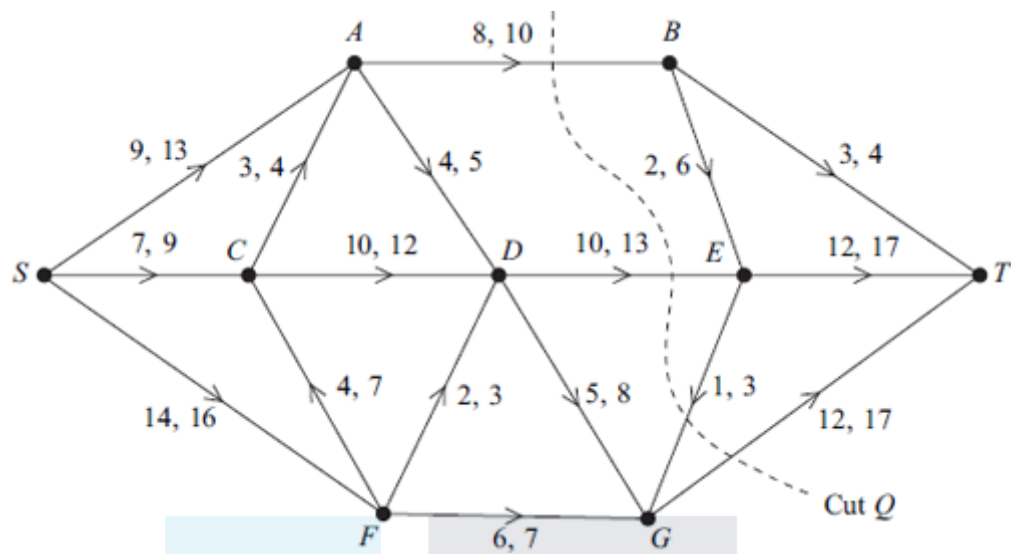
- (c) Confirm that you have a maximum flow by finding a cut of the same value. List the edges of your cut.

(2)

(Total 11 marks)

Q4.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



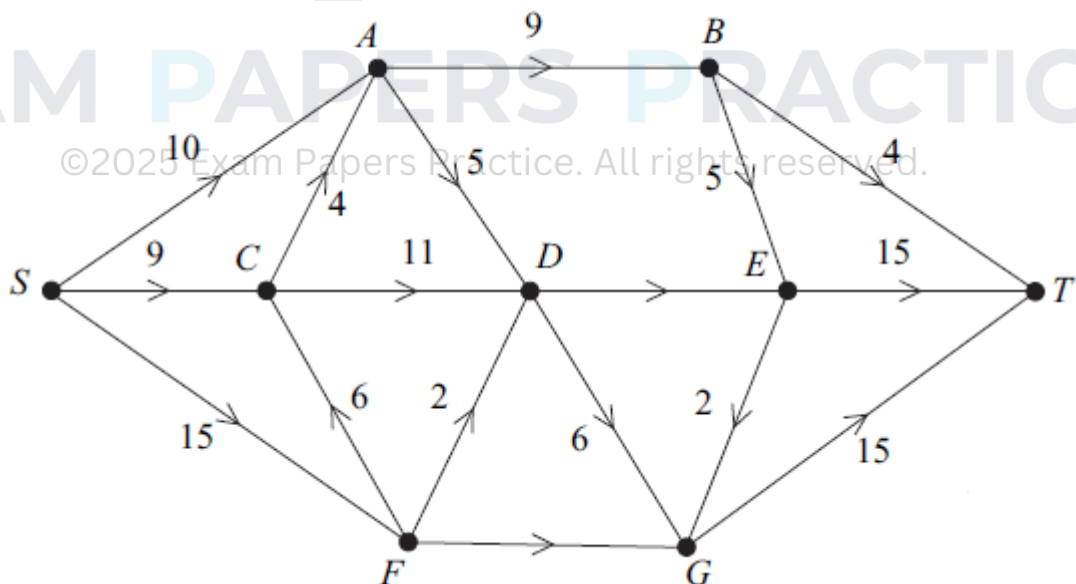
- (a) Find the value of the cut Q .

(2)

- (b) **Figure 1** shows most of the values of a feasible flow of 34 litres per second from S to T .

- (i) Insert the missing values of the flows along DE and FG on **Figure 1**.

Figure 1

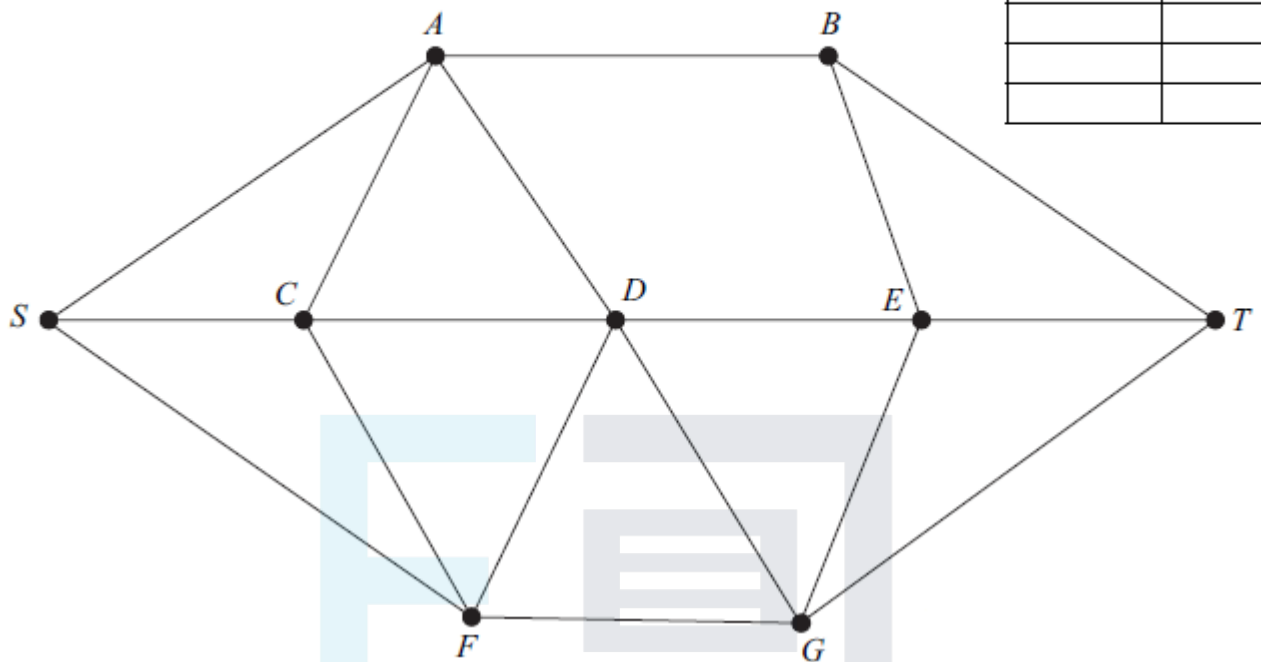


(2)

- (ii) Using this feasible flow as the initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2

Path	Extra Flow



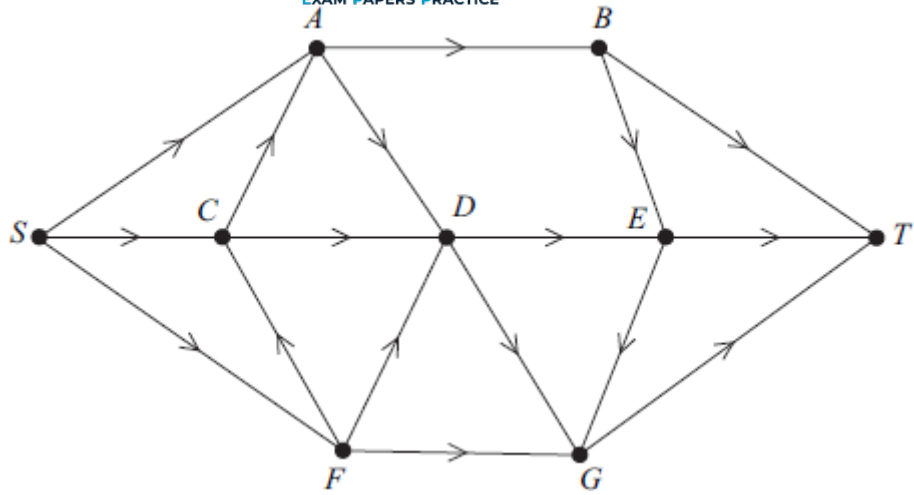
- (2)
- (iii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow-augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

- (4)
- (c) (i) State the value of the maximum flow.

©2025 Exam Papers Practice. All rights reserved. Maximum flow is _____

- (1)
- (ii) Illustrate your maximum flow on **Figure 3**.

Figure 3



(2)

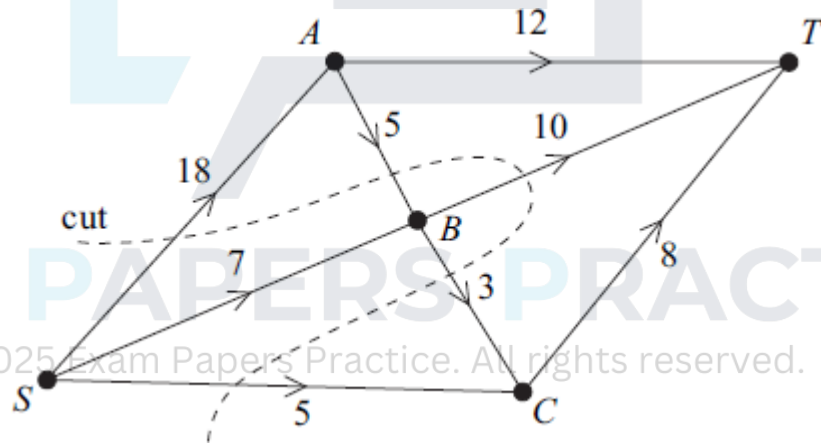
- (d) Find a cut with capacity equal to that of the maximum flow.

(1)

(Total 14 marks)

Q5.

- (a) The network shows a flow from S to T along a system of pipes, with the capacity in litres per second indicated on each edge.



- (i) Show that the value of the cut shown on the diagram is 36 .

(1)

- (ii) The cut shown on the diagram can be represented as $\{S, B\}$, $\{A, C, T\}$.

Complete the table below to give the value of each of the 8 possible cuts.

Cut		Value
$\{S\}$	$\{A, B, C, T\}$	30



$\{S, A\}$	$\{B, C, T\}$	29
$\{S, B\}$	$\{A, C, T\}$	36
$\{S, C\}$	$\{A, B, T\}$	33
$\{S, A, B\}$	$\{C, T\}$	
$\{S, A, C\}$	$\{B, T\}$	
$\{S, B, C\}$	$\{A, T\}$	
$\{S, A, B, C\}$	$\{T\}$	30

(3)

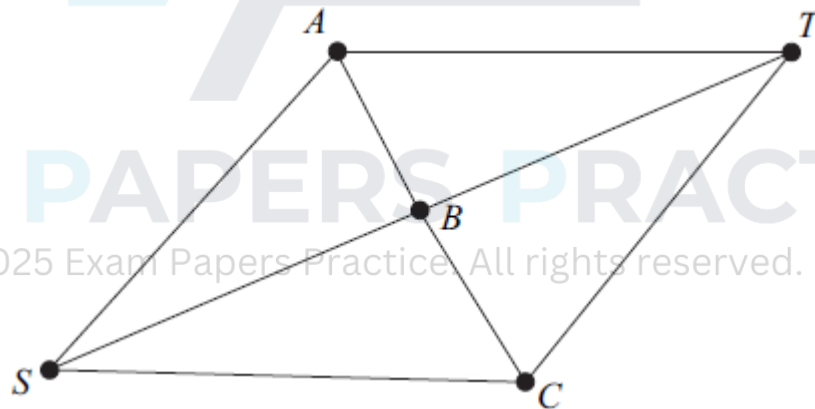
- (iii) State the value of the maximum flow through the network, giving a reason for your answer.

Maximum flow is _____

because _____

(2)

- (iv) Indicate on the diagram below a possible flow along each edge corresponding to this maximum flow.

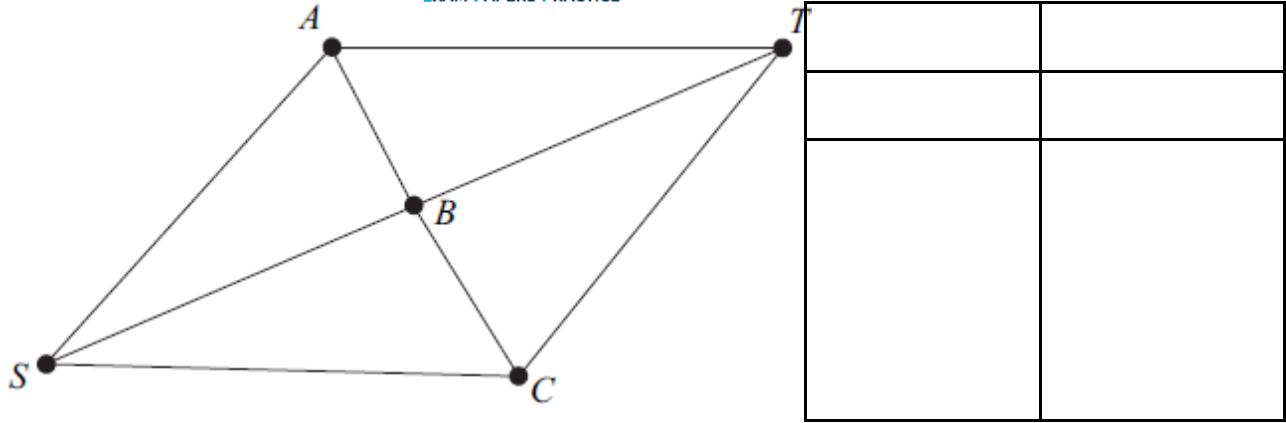


(1)

- (b) The capacities along SC and along AT are each increased by 4 litres per second.

- (i) Using your **values from part (a)(iv) as the initial flow**, indicate potential increases and decreases on the diagram below and use the labelling procedure to find the new maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the diagram.

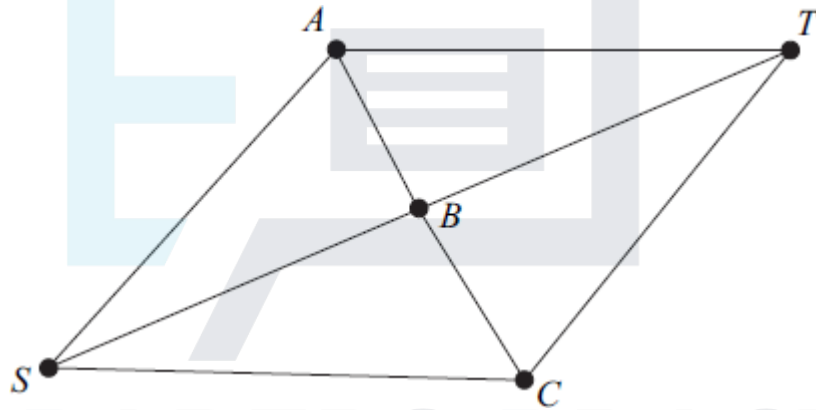
Path	Additional Flow



(6)

- (ii) Use your results from part (b)(i) to illustrate the flow along each edge that gives this new maximum flow, and state the value of the new maximum flow.

New maximum flow is _____



(3)

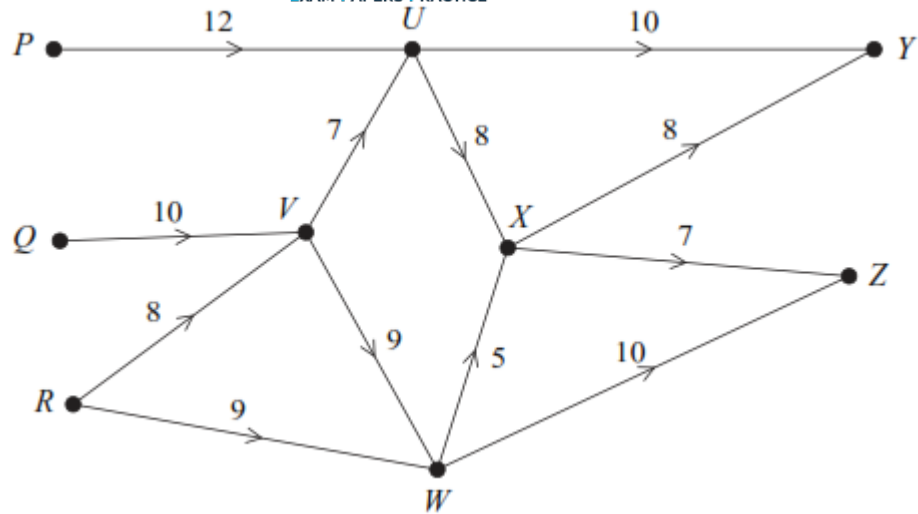
(Total 16 marks)

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Q6.

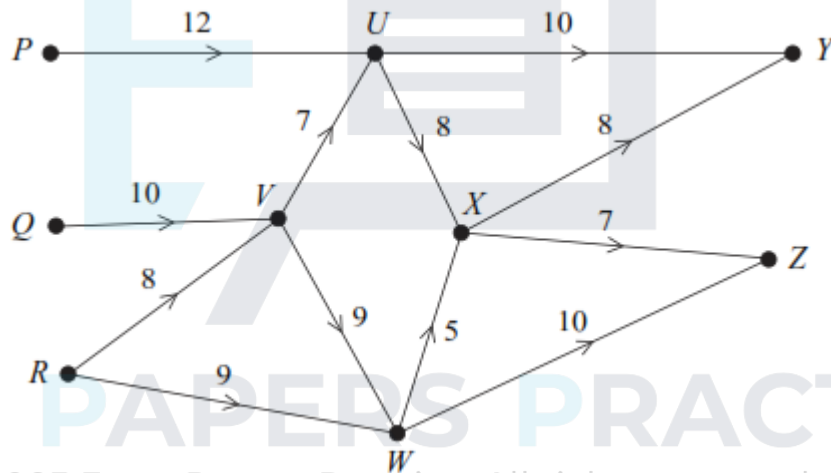
A retail company has warehouses at P , Q and R , and goods are to be transported to retail outlets at Y and Z . There are also retaining depots at U , V , W and X .

The possible routes with the capacities along each edge, in van loads per week, are shown in the following diagram.



- (a) On **Figure 1**, add a super-source, S , and a super-sink, T , and appropriate edges so as to produce a directed network with a single source and a single sink. Indicate the capacity of each edge that you have added.

Figure 1



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(2)

- (b) On **Figure 2**, write down the maximum flows along the routes $SPUYT$ and $SRVWZT$.

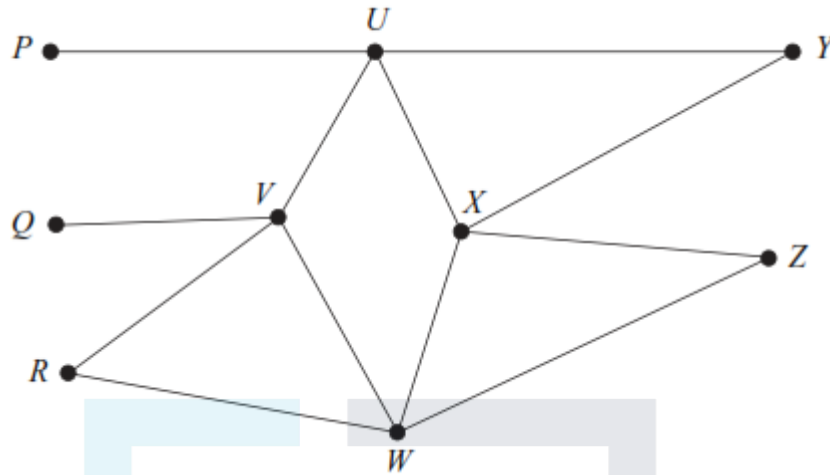
Figure 2

Route	Flow
$SPUYT$	
$SRVWZT$	

(2)

- (c) (i) On **Figure 3**, add the vertices S and T and the edges connecting S and T to the network. Using the maximum flows along the routes $SPUYT$ and $SRVWZT$ found in part (b) as the initial flow, indicate the potential increases and decreases of the flow on each edge of these routes.

Figure 3

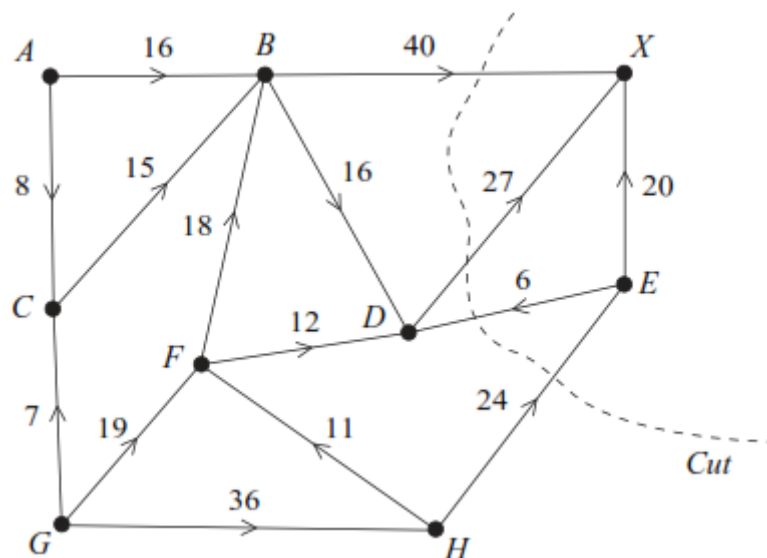


- (2)
- (ii) Use flow augmentation to find the maximum flow from S to T . You should indicate any flow-augmenting routes on **Figure 2** and modify the potential increases and decreases of the flow on **Figure 3**.
- (4)
- (d) Find a cut with value equal to the maximum flow.

(1)
(Total 11 marks)

Q7.

The network shows the evacuation routes along corridors in a college, from two teaching areas to the exit, in case of a fire alarm sounding.



The two teaching areas are at A and G and the exit is at X .

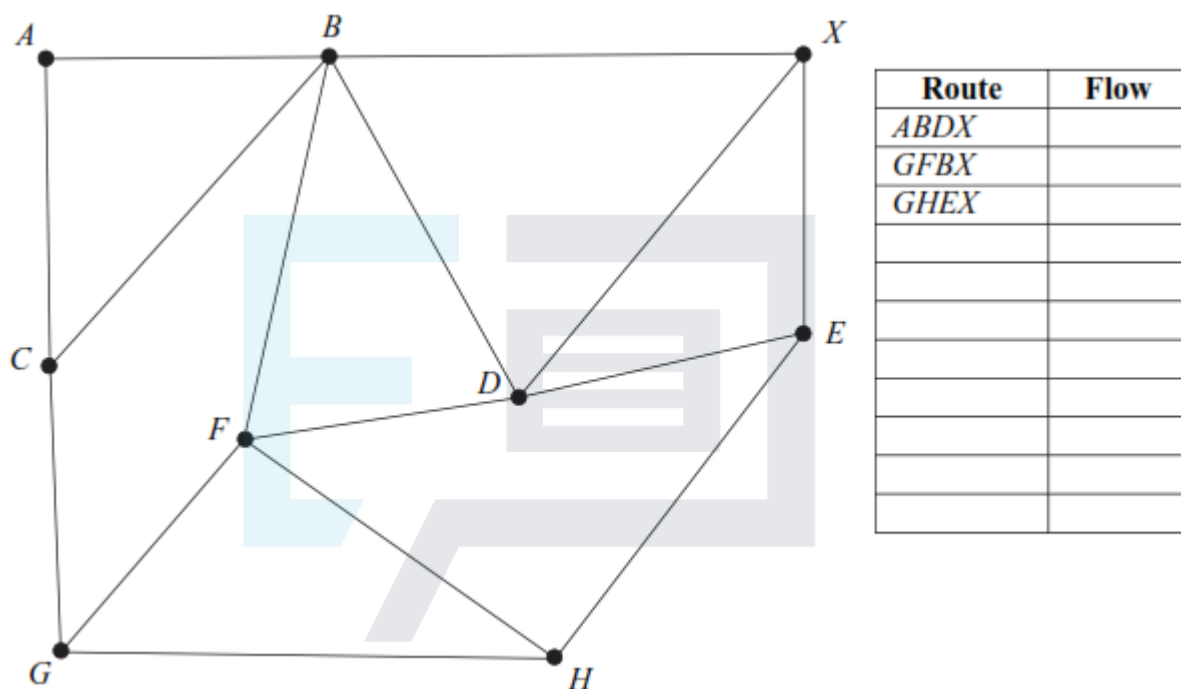
The number on each edge represents the maximum number of people that can travel along a particular corridor in one minute.

- (a) Find the value of the cut shown on the diagram.

(1)

- (b) Find the maximum flow along each of the routes $ABDX$, $GFBX$ and $GHEX$ and enter their values in the table on **Figure 1**.

Figure 1

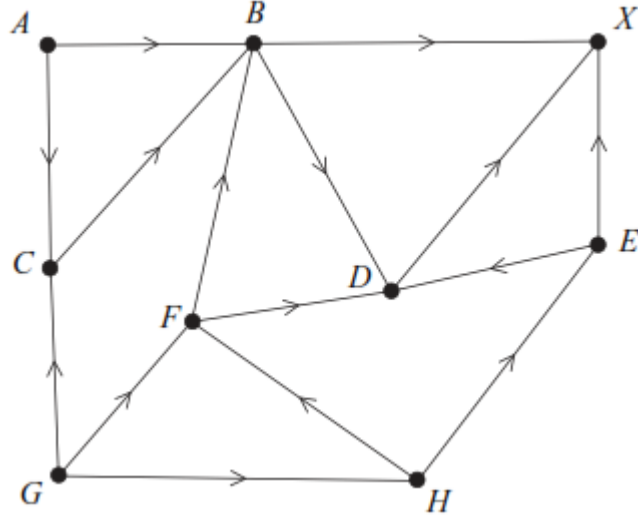


- (c) (i) Taking your answers to part (b) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting routes in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2



Maximum flow is _____ people per minute.

(2)

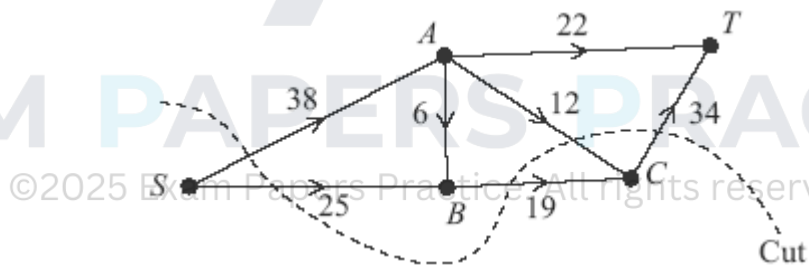
- (d) During one particular fire drill, there is an obstruction allowing no more than 45 people per minute to pass through vertex B . State the maximum number of people that can move through the network per minute during this fire drill.

(2)

(Total 13 marks)

Q8.

- (a) The network shows a flow from S to T along a system of pipes, with the capacity, in litres per minute, indicated on each edge.



- (i) Show that the value of the cut shown on the diagram is 97.

(1)

- (ii) The cut shown on the diagram can be represented as $\{S, C\}, \{A, B, T\}$.

Complete the table on **Figure 1**, giving the value of each of the 8 possible cuts.

Figure 1

Cut		Value
$\{S\}$	$\{A, B, C, T\}$	63
$\{S, A\}$	$\{B, C, T\}$	

$\{S, B\}$	$\{A, C, T\}$	
$\{S, C\}$	$\{A, B, T\}$	97
$\{S, A, B\}$	$\{C, T\}$	53
$\{S, A, C\}$	$\{B, T\}$	87
$\{S, B, C\}$	$\{A, T\}$	
$\{S, A, B, C\}$	$\{T\}$	

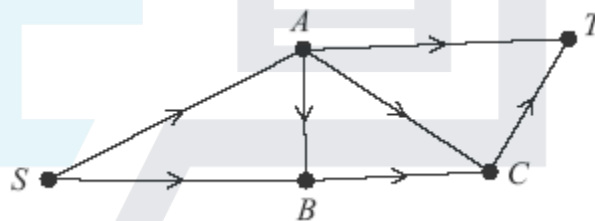
(4)

- (iii) State the value of the maximum flow through the network, giving a reason for your answer.

(2)

- (iv) Indicate on **Figure 2** a possible flow along each edge corresponding to this maximum flow.

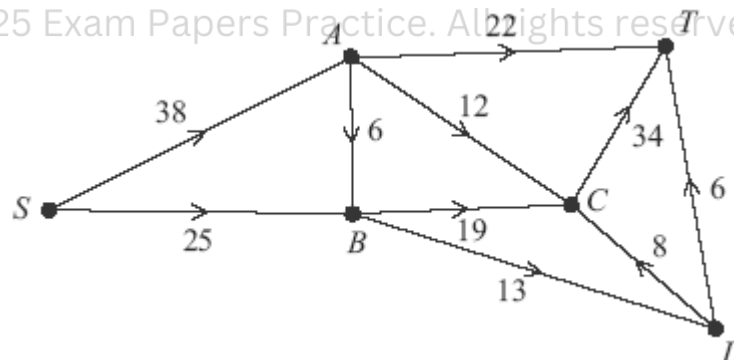
Figure 2



(2)

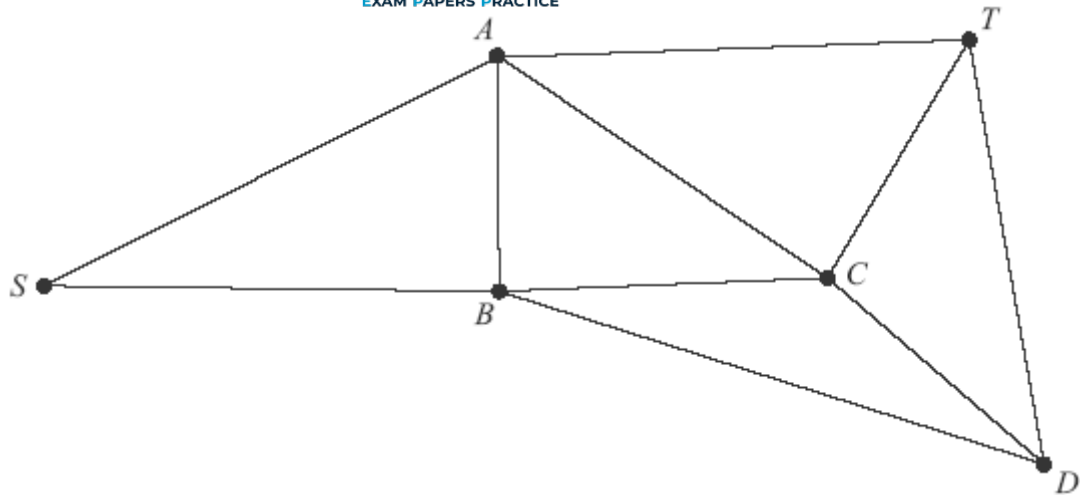
- (b) Extra pipes, BD , CD and DT , are added to form a new system, with the capacity, in litres per minute, indicated on each edge of the network below.

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- (i) Taking your values from Figure 2 as the initial flow, use the labelling procedure on Figure 3 to find the new maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 3

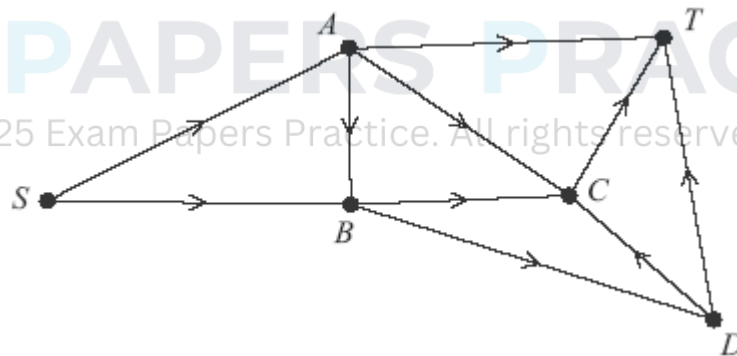


Path	Additional flow

(4)

- (ii) State the value of the new maximum flow, and, on **Figure 4**, indicate a possible flow along each edge corresponding to this maximum flow.

Figure 4



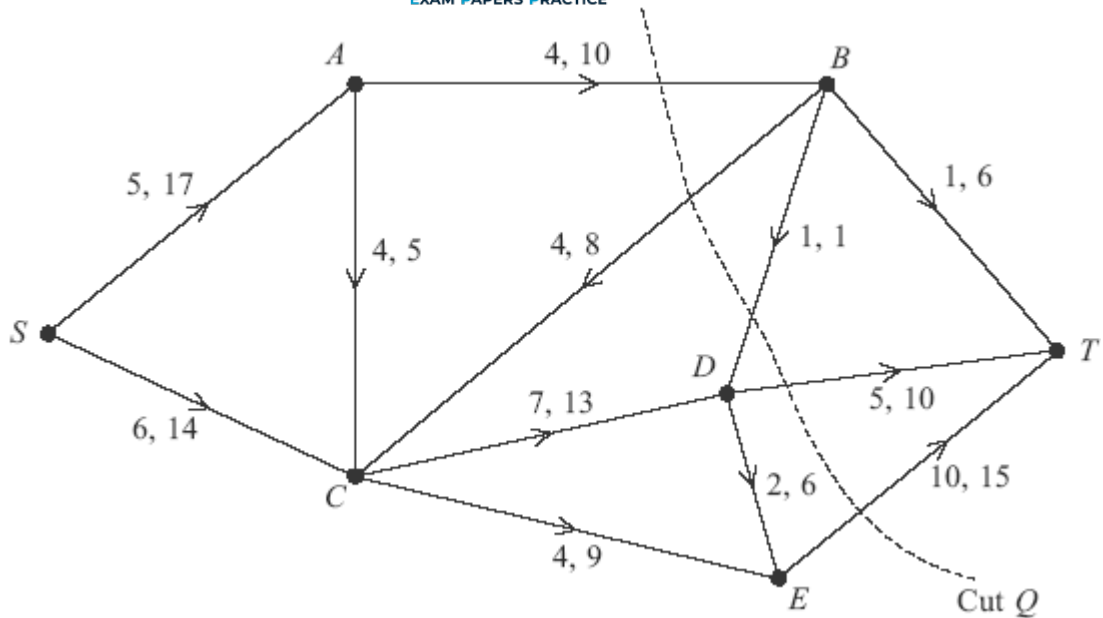
New maximum flow is _____

(2)

(Total 15 marks)

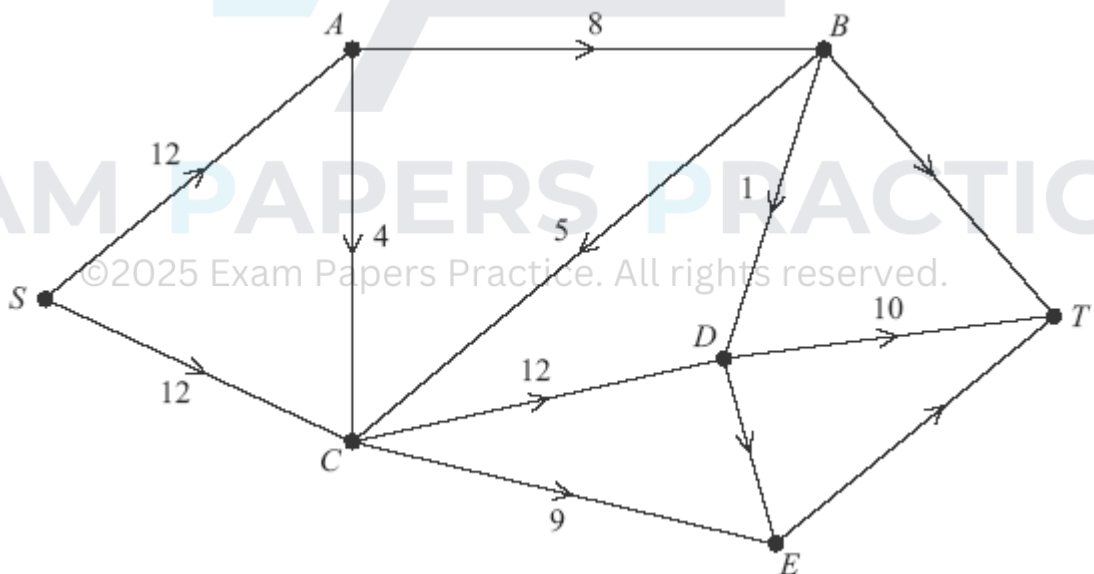
Q9.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per minute.



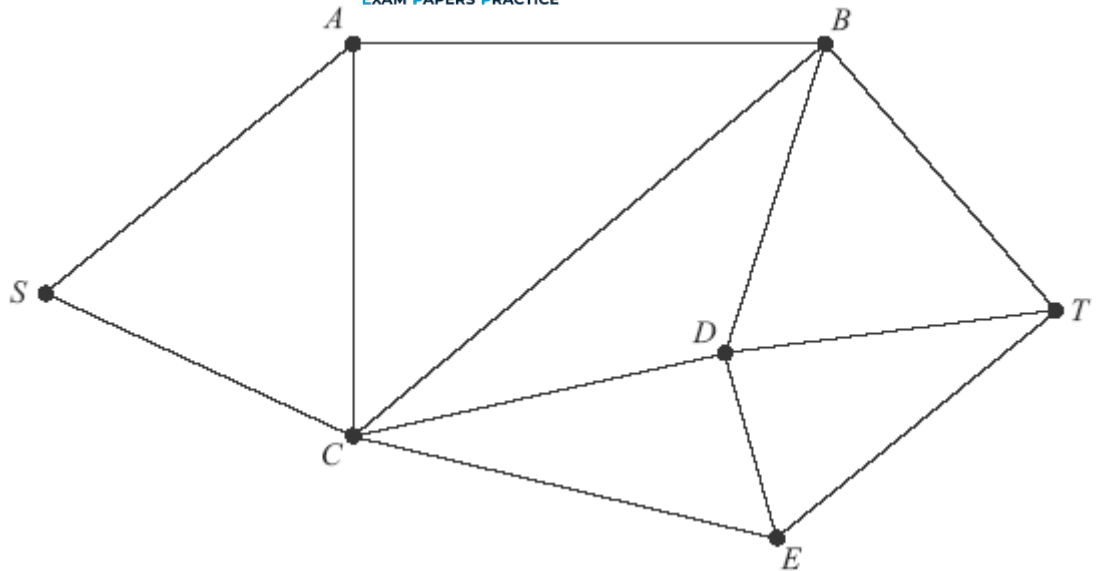
- (a) Find the value of the cut Q . (2)
- (b) **Figure 1** below shows a partially completed diagram for a feasible flow of 24 litres per minute from S to T . Indicate, on **Figure 1**, the flows along the edges BT , DE and ET .

Figure 1



- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2** below. (2)

Figure 2



Path	Flow

(2)

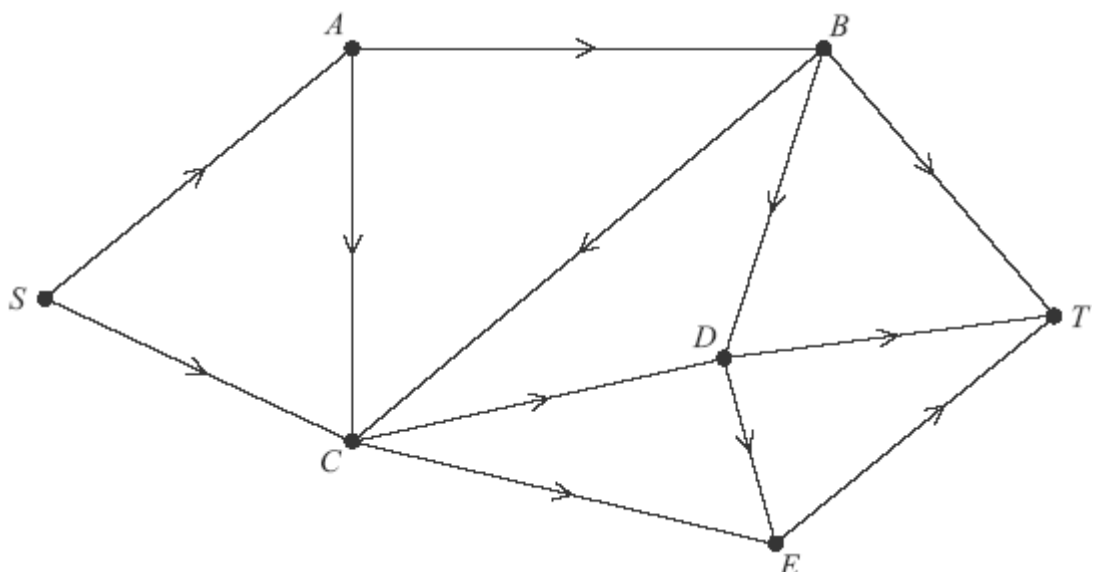
- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T .

You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (iii) Illustrate the maximum flow on **Figure 3** below.

Figure 3



(2)

- (d) Find a cut with value equal to that of the maximum flow.

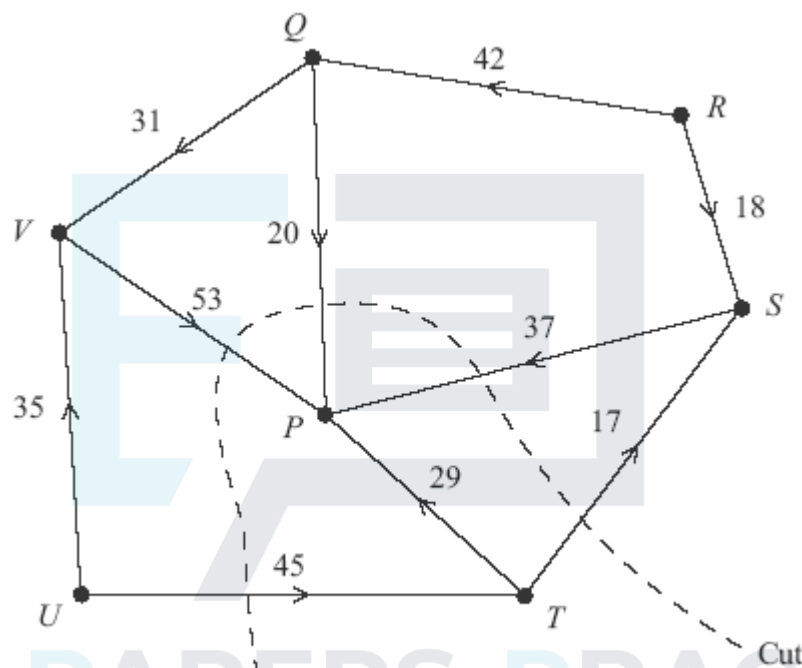
You may wish to show the cut on the network above.

(1)

(Total 14 marks)

Q10.

The network shows the routes along corridors from two arrival gates to the passport control area, P , in a small airport. The number on each edge represents the maximum number of passengers that can travel along a particular corridor in one minute.



- (a) State the vertices that represent the arrival gates.

(1)

- (b) Find the value of the cut shown on the network.

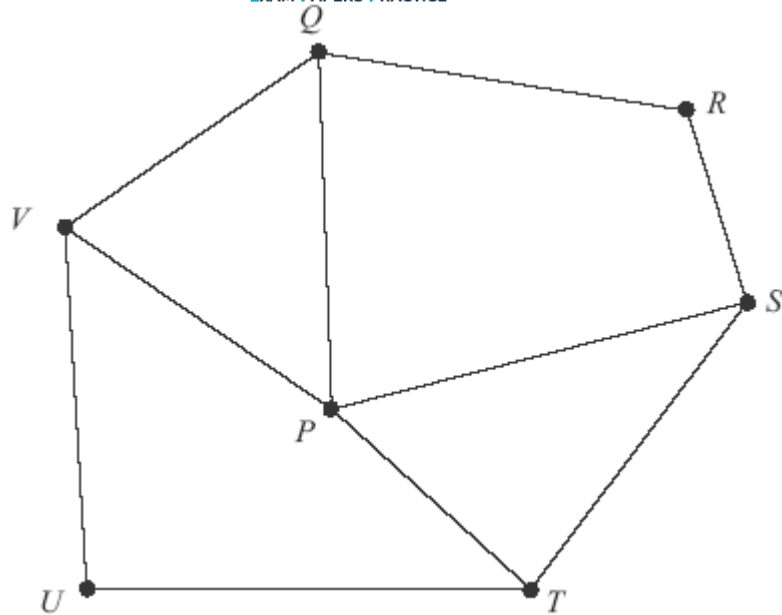
(1)

- (c) State the maximum flow along each of the routes $UTSP$ and $RQVP$.

(2)

- (d) (i) Taking your answers to part (c) as the initial flow, use the labelling procedure on **Figure 1** to find the maximum flow through the network. You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

Figure 1



Route	Value of Flow
<i>UTSP</i>	
<i>RQVP</i>	

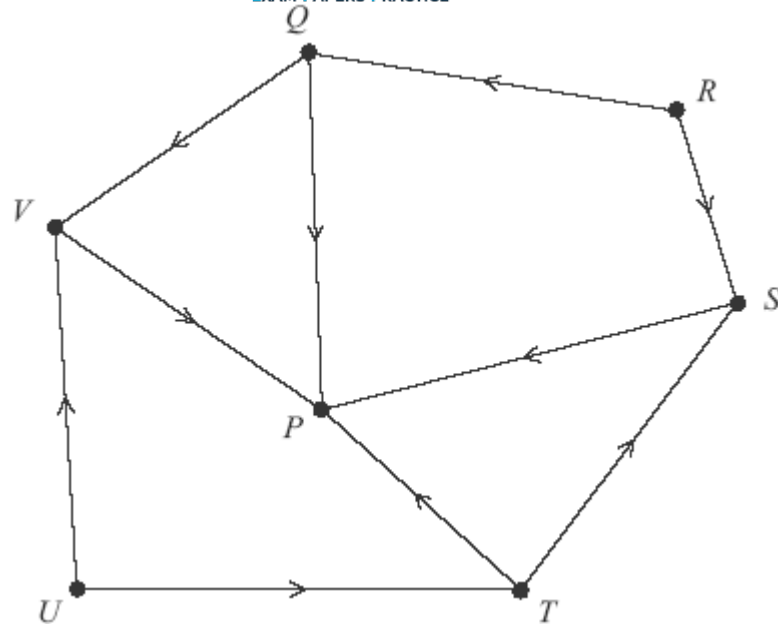
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(6)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

Figure 2

Maximum flow is _____ passengers per minute



(2)

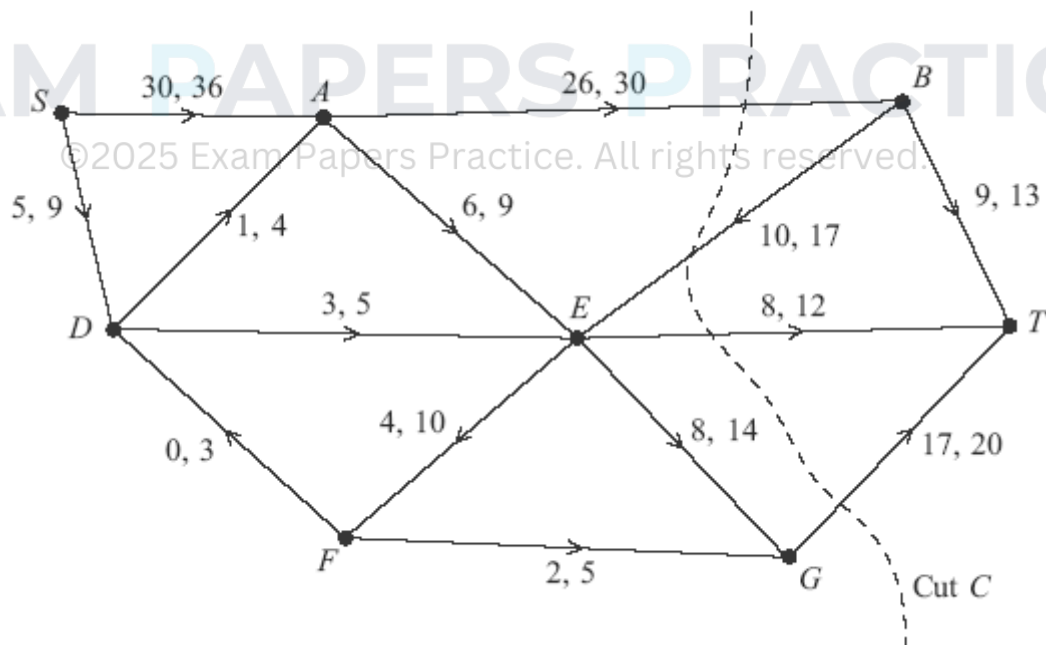
- (e) On a particular day, there is an obstruction allowing no more than 50 passengers per minute to pass through vertex V . State the maximum number of passengers that can move through the network per minute on this particular day.

(2)

(Total 14 marks)

Q11.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



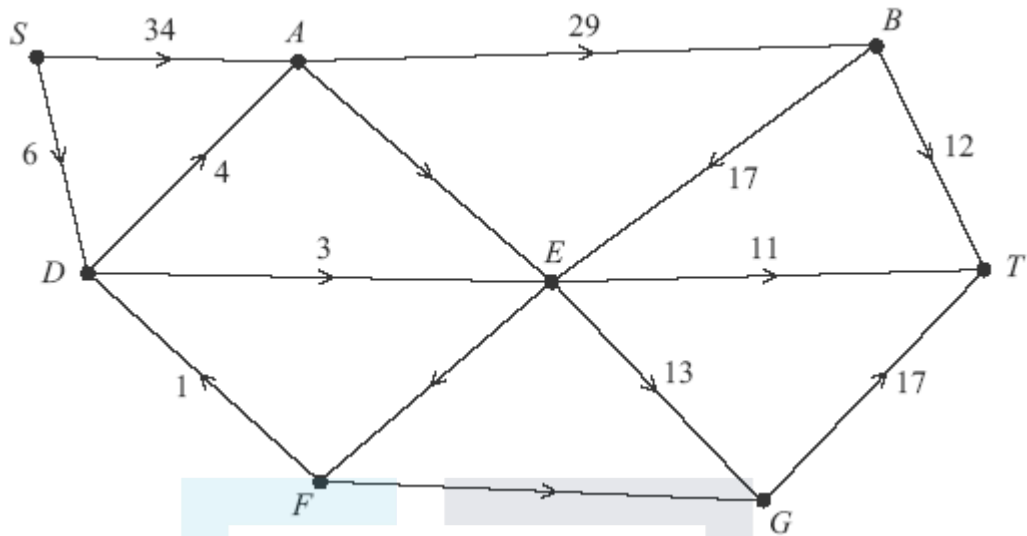
- (a) Find the value of the cut C .

(2)

- (b) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 40 litres

per second from S to T . Indicate, on **Figure 1**, the flows along the edges AE , EF and FG .

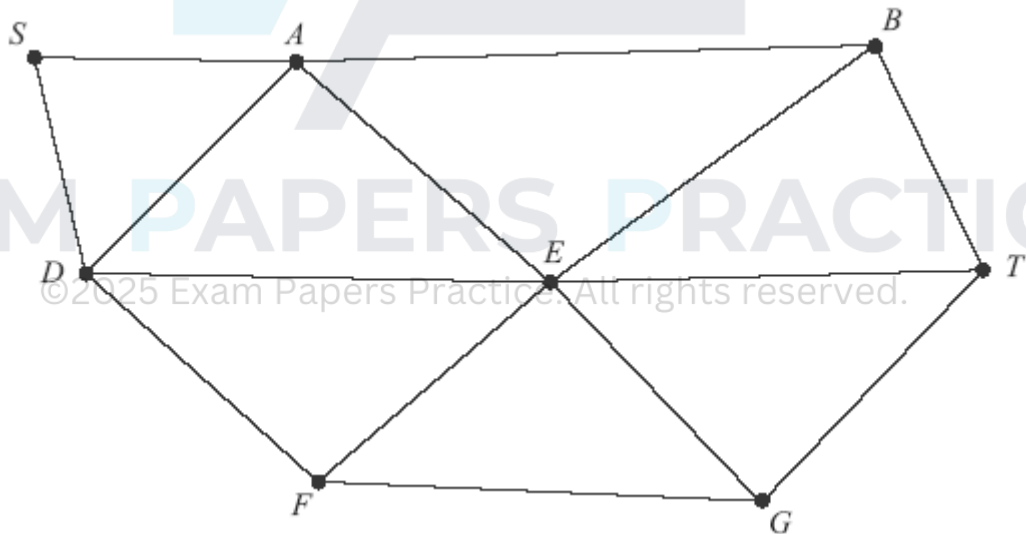
Figure 1



(3)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Extra Flow

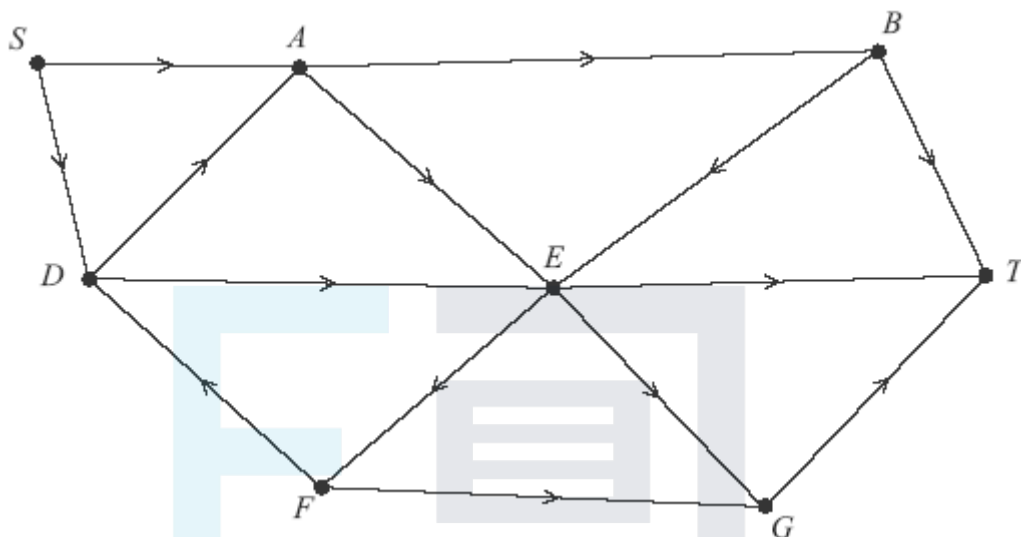
(3)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(4)

- (d) Illustrate the maximum flow on **Figure 3**.

Figure 3



(2)

- (e) Find a cut with value equal to that of the maximum flow.

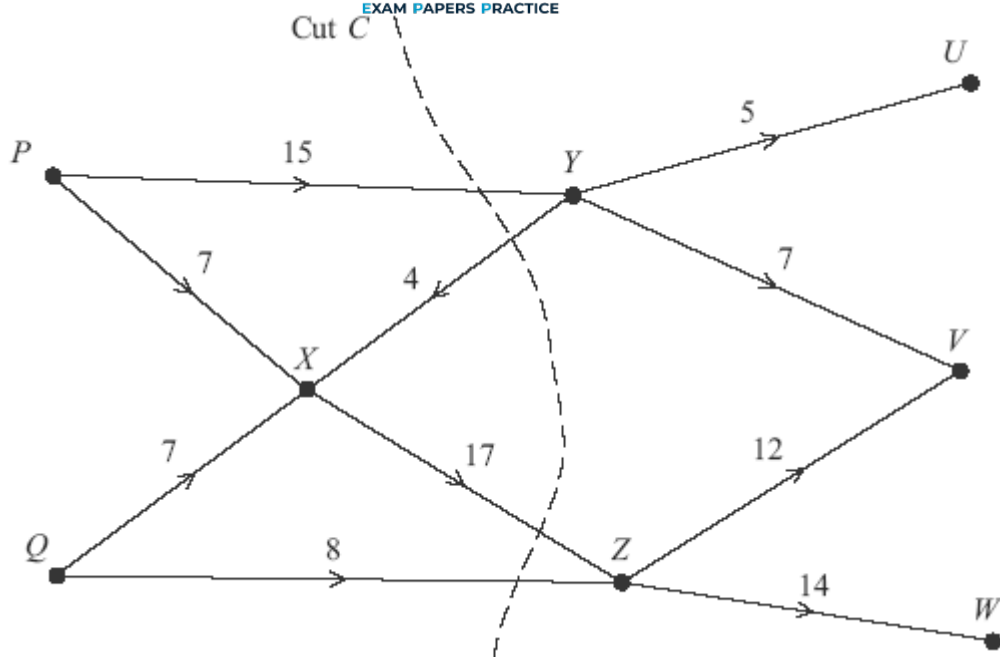
(2)

(Total 16 marks)

Q12.

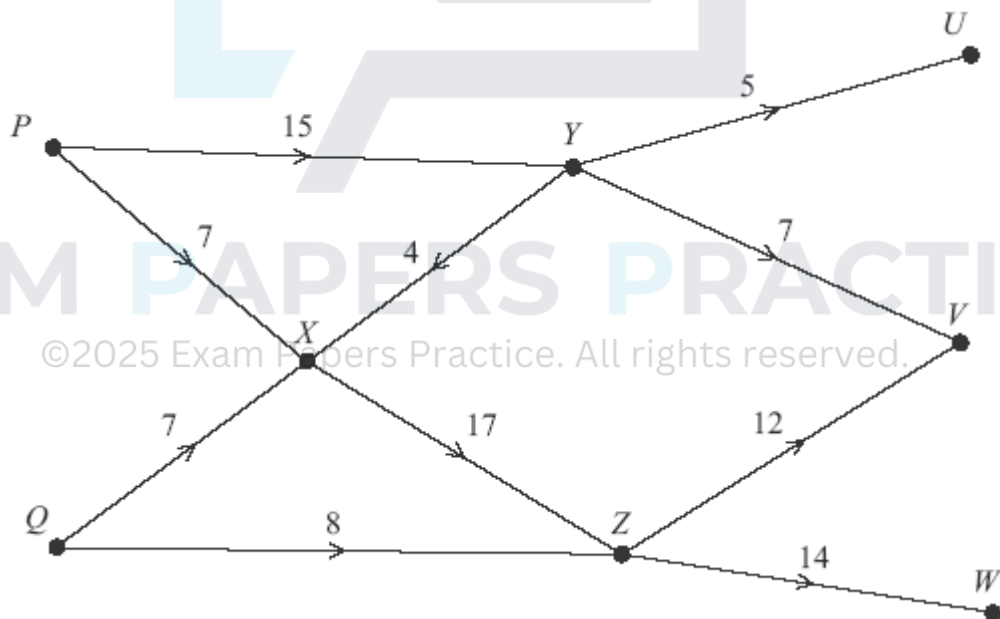
Water has to be transferred from two mountain lakes P and Q to three urban reservoirs U , V and W . There are pumping stations at X , Y and Z .

The possible routes with the capacities along each edge, in millions of litres per hour, are shown in the following diagram.



- (a) On **Figure 1**, add a super-source, S , and a super-sink, T , and appropriate edges so as to produce a directed network with a single source and a single sink. Indicate the capacity of each of the edges you have added.

Figure 1



- (b) (i) Find the value of the cut C .
(ii) State what can be deduced about the maximum flow from S to T .
(c) On **Figure 2**, write down the maximum flows along the routes $SQZWT$ and $SPYXZVT$.

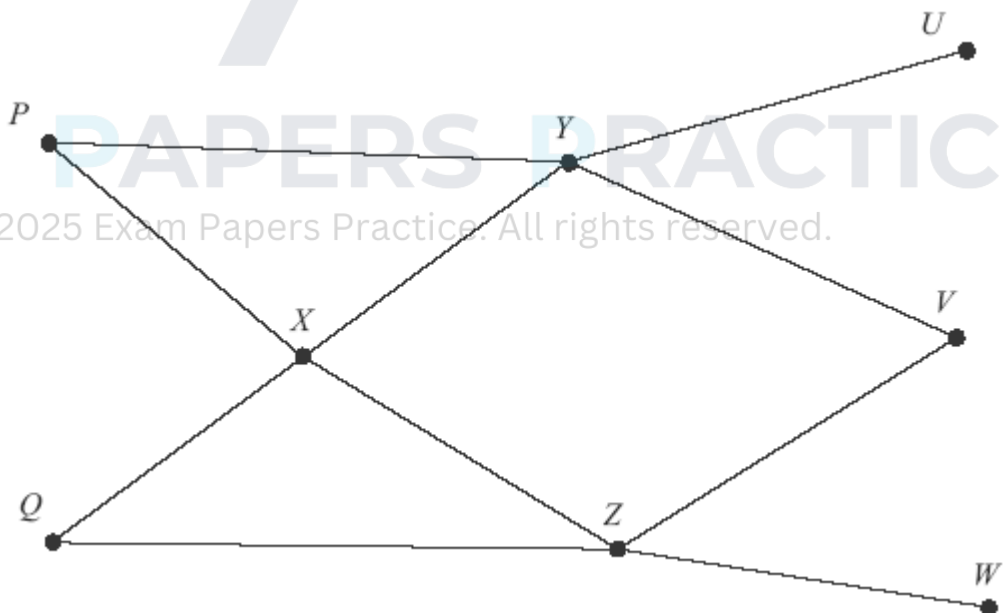
Figure 2

Route	Flow
<i>SQZWT</i>	
<i>SPYXZVT</i>	

(2)

- (d) (i) On **Figure 3**, add the vertices *S* and *T* and the edges connecting *S* and *T* to the network. Using the maximum flows along the routes *SQZWT* and *SPYXZVT* found in part (c) as the initial flow, indicate the potential increases and decreases of flow on each edge.

Figure 3



(2)

- (ii) Use flow augmentation to find the maximum flow from *S* to *T*. You should indicate any flow augmenting paths on **Figure 2** and modify the potential increases and decreases of the flow on **Figure 3**.

(4)

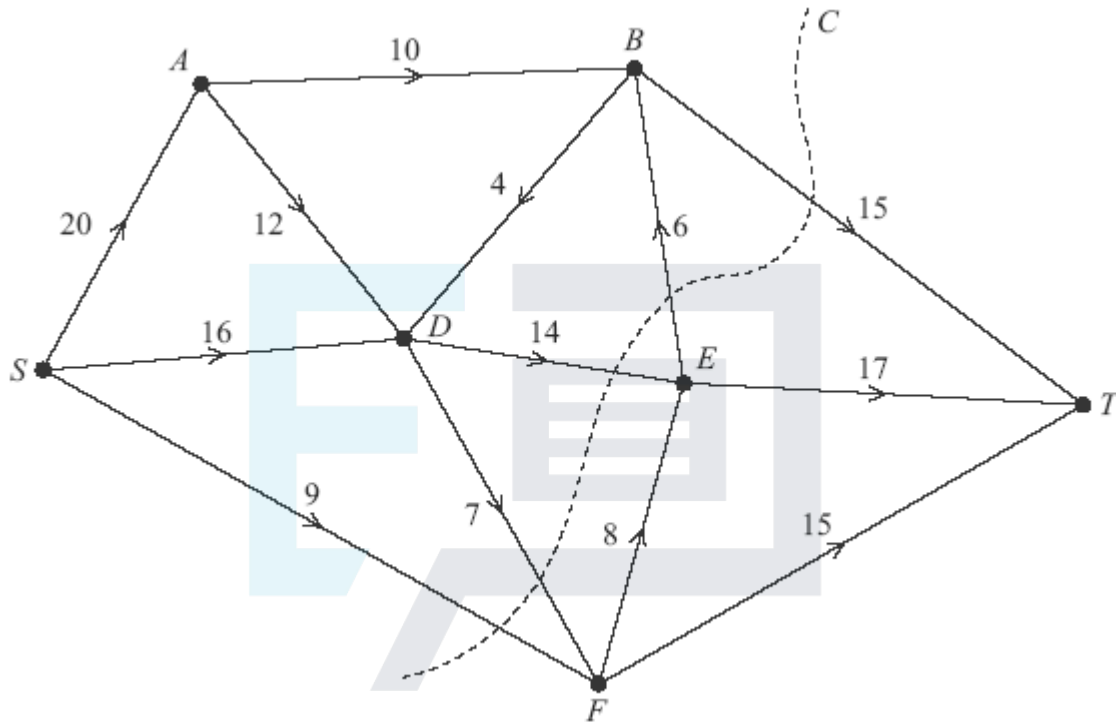
- (e) State the value of the flow from *Y* to *X* in millions of litres per hour when the

maximum flow is achieved.

(1)
(Total 13 marks)

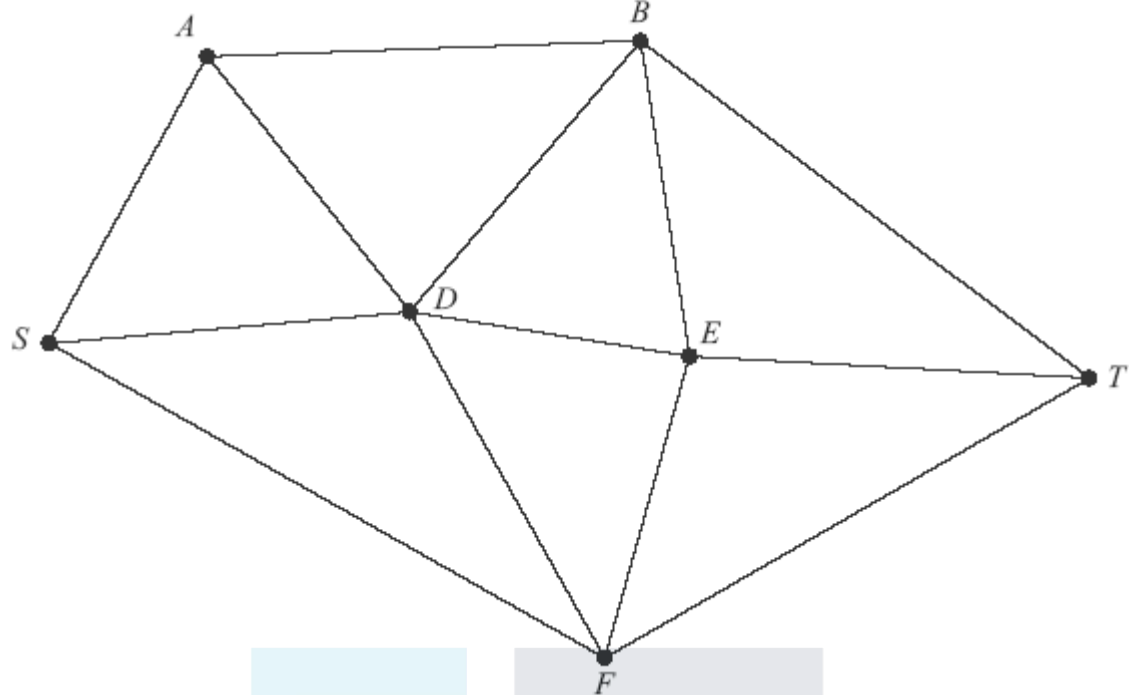
Q13.

The diagram shows a network of pipelines through which oil can travel. The oil field is at S , the refinery is at T and the other vertices are intermediate stations. The weights on the edges show the capacities in millions of barrels per hour that can flow through each pipeline.



- (a) (i) Find the value of the cut marked C on the diagram. (1)
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- (ii) Hence make a deduction about the maximum flow of oil through the network. (2)
- (b) State the maximum possible flows along the routes $SABT$, $SDET$ and SFT . (2)
- (c) (i) Taking your answer to part (b) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow from S to T . Record your routes and flows in the table provided and show the augmented flows on the network diagram.

Figure 1



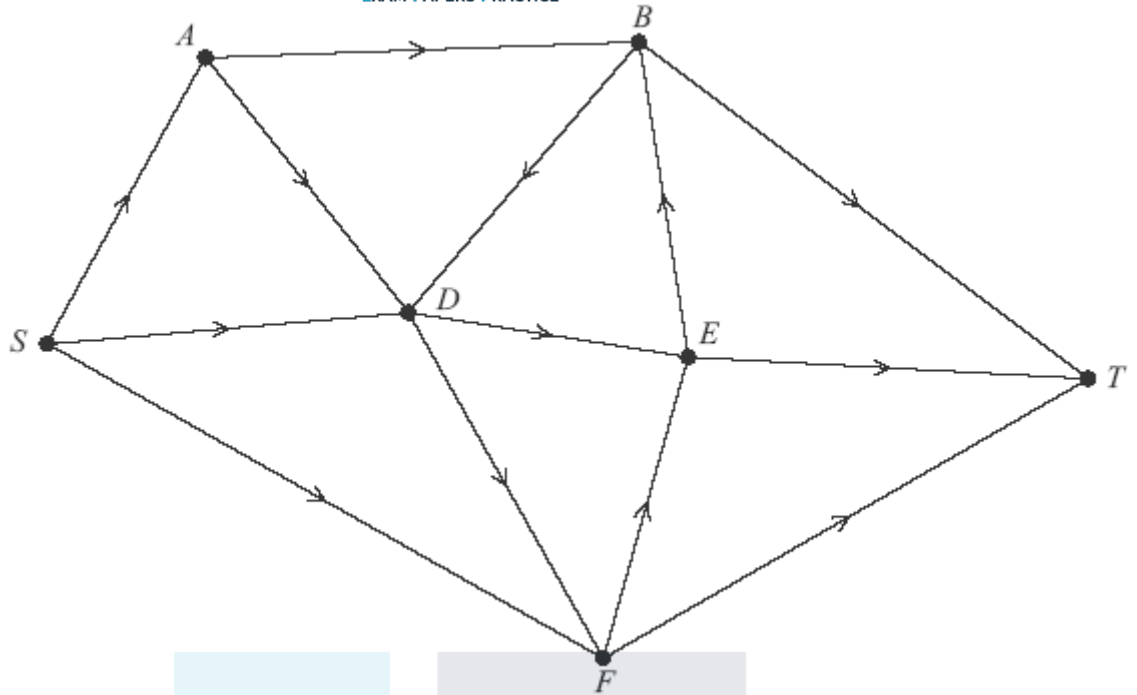
Route	Flow
<i>SABT</i> <i>SDET</i> <i>SFT</i>	

(6)

- (ii) State the value of the maximum flow, and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

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Figure 2



(2)

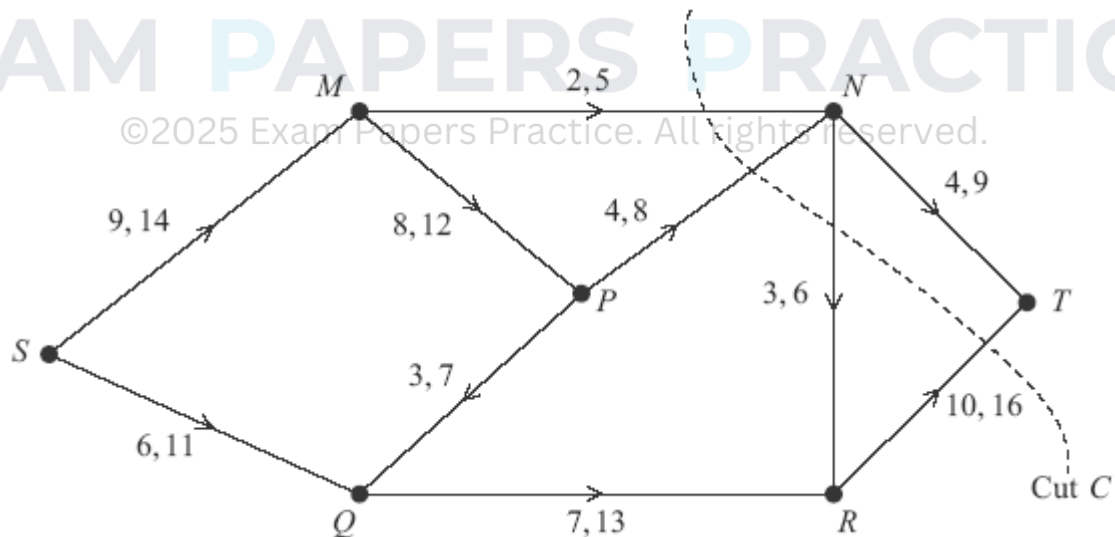
(iii) Prove that your flow in part (c)(ii) is a maximum.

(2)

(Total 15 marks)

Q14.

The network shows a system of pipes with the lower and upper capacities for each pipe in litres per second.



(a) (i) Find the value of the cut C .

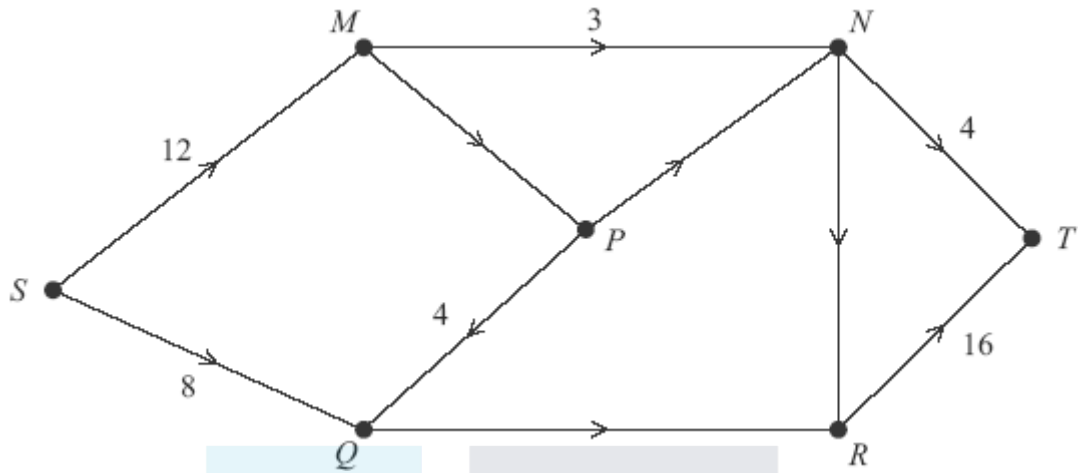
(1)

(ii) State what can be deduced about the maximum flow from S to T .

(1)

- (b) **Figure 1** shows a partially completed diagram for a feasible flow of 20 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MP , PN , QR and NR .

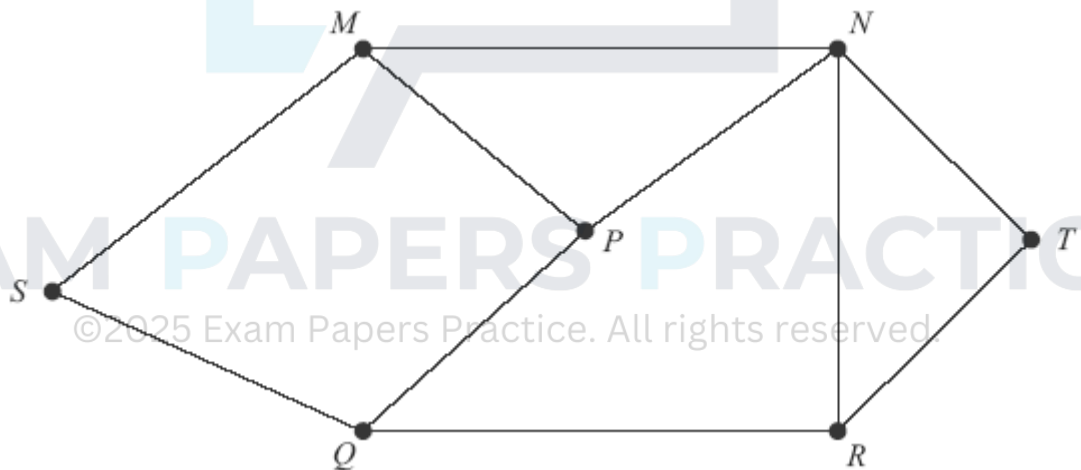
Figure 1



(4)

- (c) (i) Taking your answer from part (b) as an initial flow, indicate potential increases and decreases of the flow along each edge on **Figure 2**.

Figure 2



Path	Flow

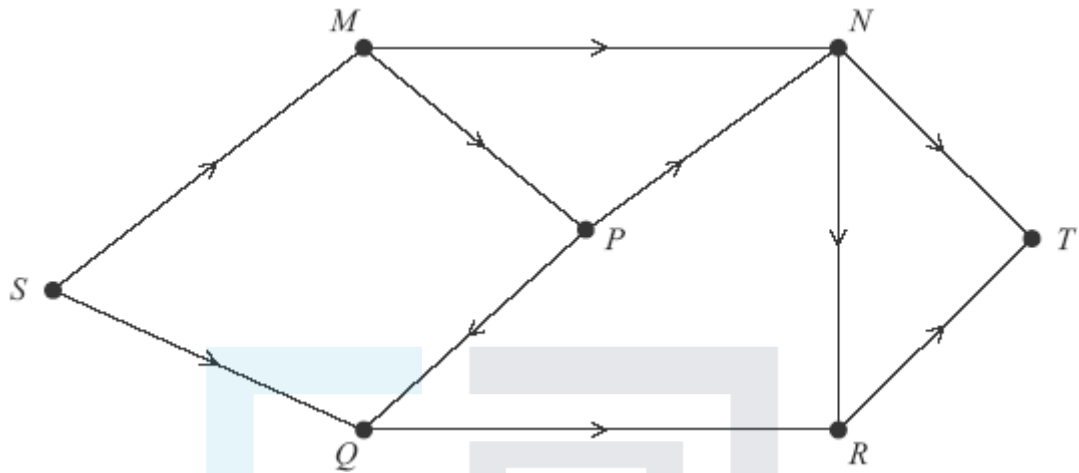
(2)

- (ii) Use flow augmentation on **Figure 2** to find the maximum flow from S to T . You should indicate any flow augmenting paths in the table and modify the potential increases and decreases of the flow on the network.

(5)

- (iii) Illustrate the maximum flow on **Figure 3**.

Figure 3

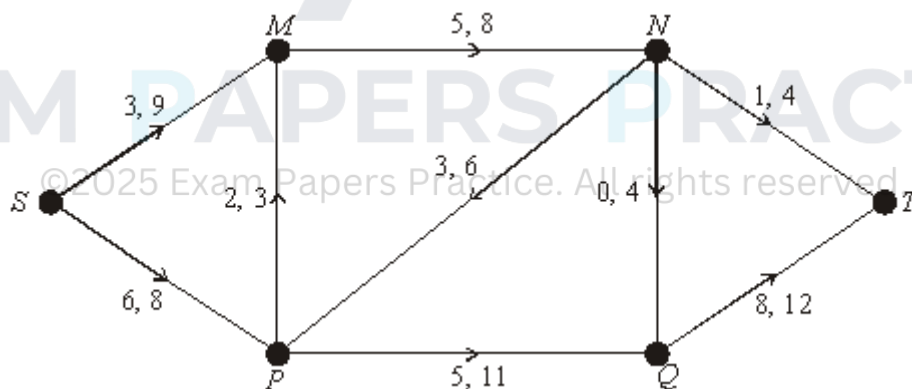


(2)

(Total 15 marks)

Q15.

The network shows a system of pipes, with the lower and upper capacities for each pipe in litres per second.



- (a) **Figure 1**, below, shows a partially completed diagram for a feasible flow of 10 litres per second from S to T . Indicate, on **Figure 1**, the flows along the edges MN , PQ , NP and NT .

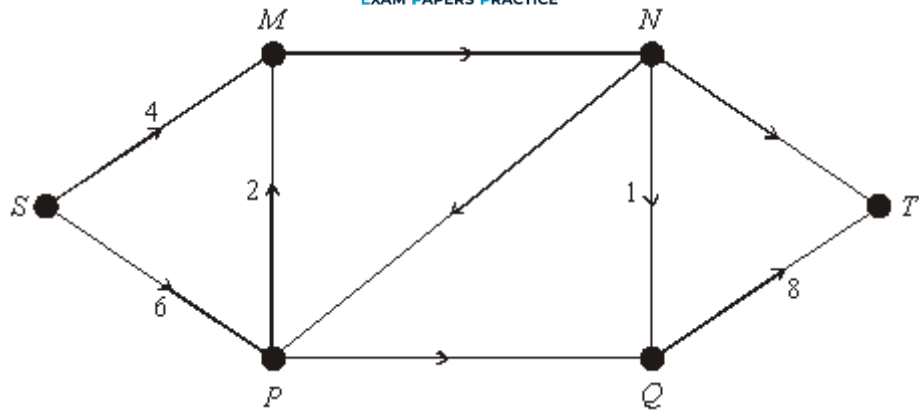
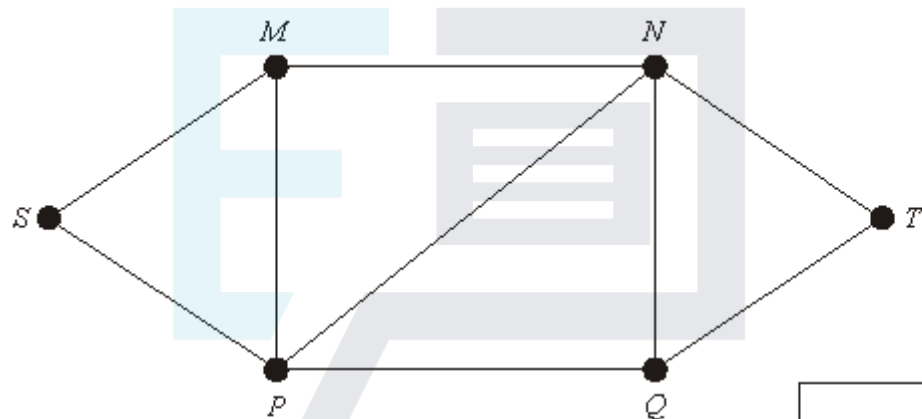


Figure 1

(4)

- (b) (i) Taking your answer from part (a) as an initial flow, use flow augmentation on **Figure 2** to find the maximum flow from S to T .



Route	Extra flow

Figure 2

(6)

- (ii) State the value of the maximum flow and illustrate this flow on **Figure 3**.

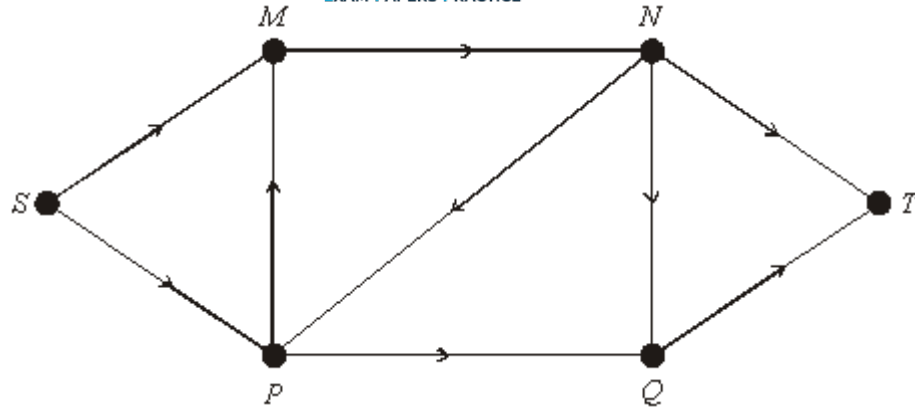


Figure 3

(2)

- (c) Find a cut with capacity equal to that of the maximum flow.

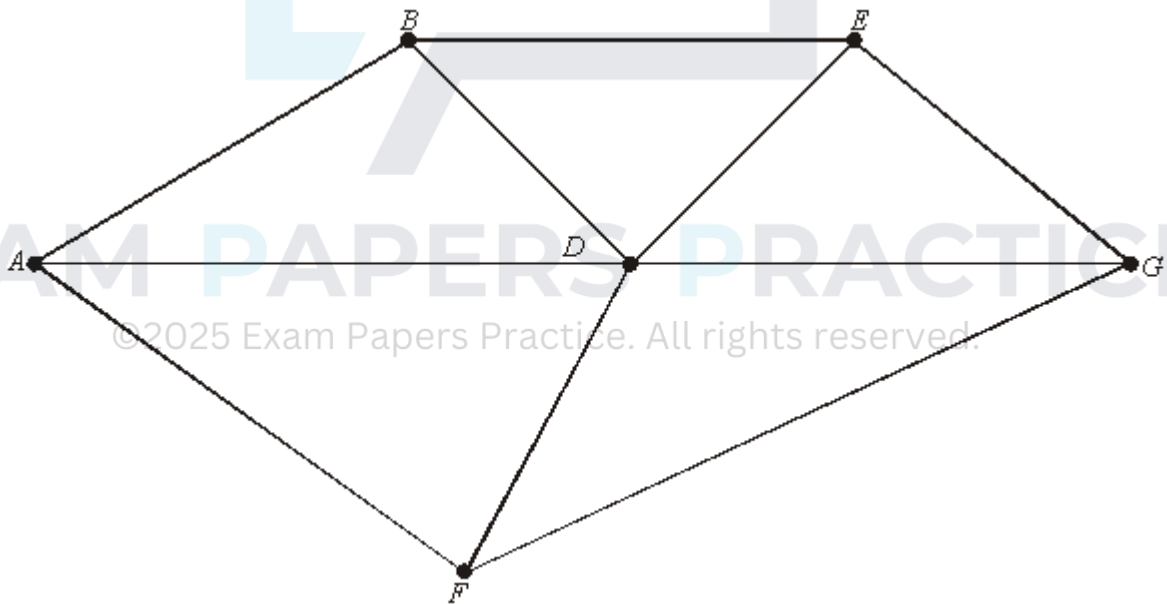
(2)

(Total 14 marks)

Q16.

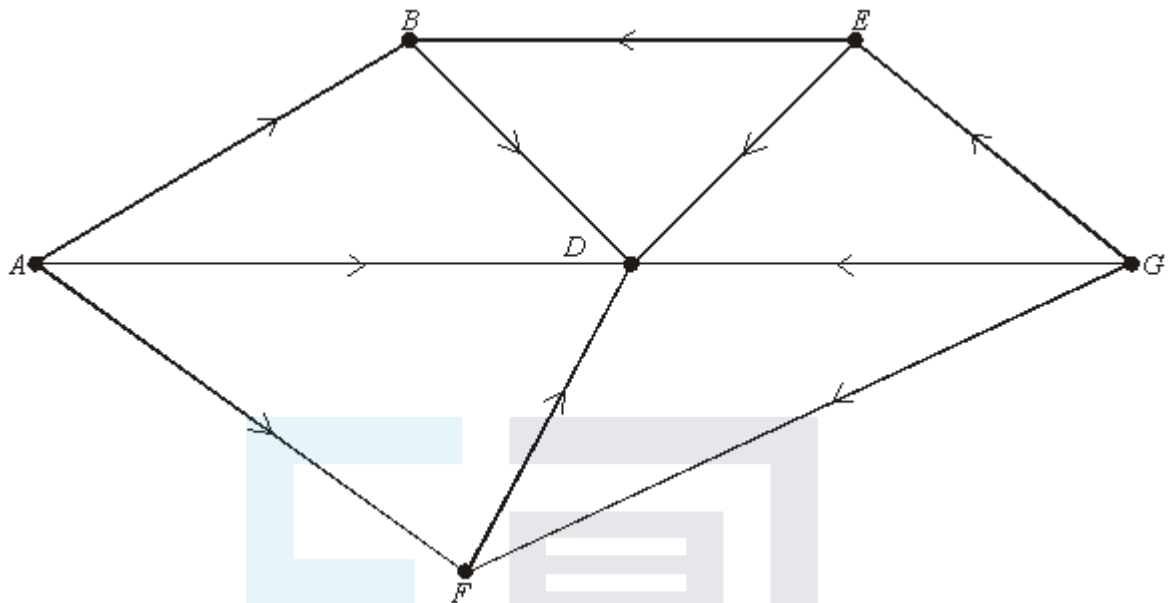
[Figures 1 and 2, below, are provided for use in this question.]

Figure 1

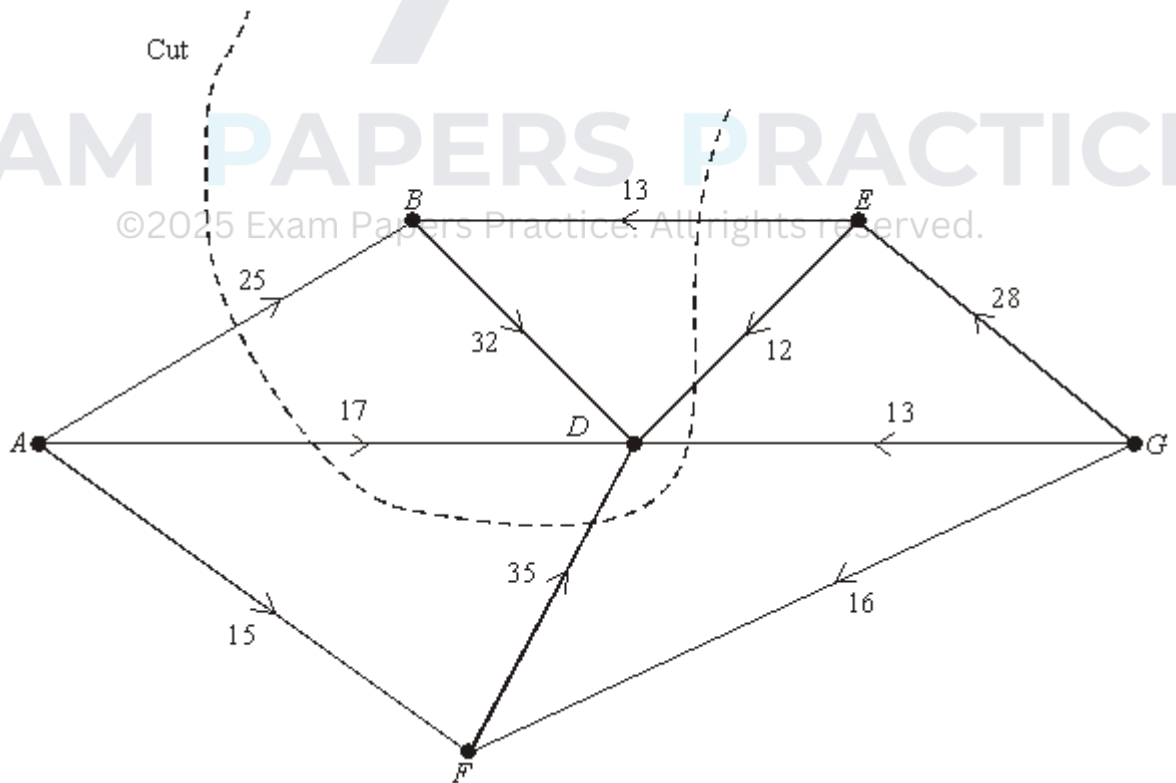


Route	Flow
ABC	
GED	

Figure 2



The network shows the routes along corridors from the playgrounds A and G to the assembly hall in a school. The number on each edge represents the maximum number of pupils that can travel along the corridor in one minute.



- (a) State the vertex that represents the assembly hall.

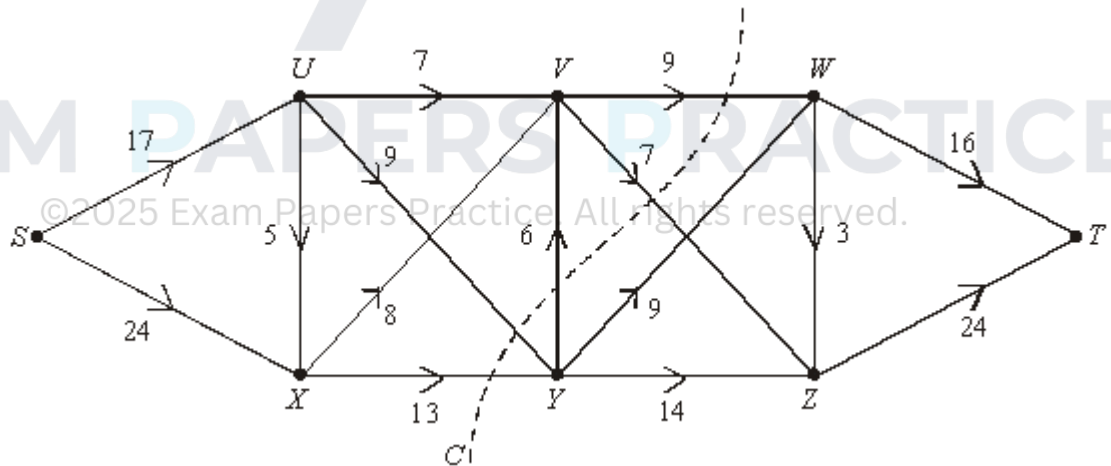
(1)

- (b) Find the value of the cut shown on the diagram. (1)
- (c) State the maximum flow along the routes ABD and GED . (2)
- (d) (i) Taking your answers to part (c) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow through the network. (6)
- (ii) State the value of the maximum flow and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow. (2)
- (iii) Verify that your flow is a maximum flow by finding a cut of the same value. (2)
- (e) On a particular day, there is an obstruction allowing no more than 15 pupils per minute to pass through vertex E . State the maximum number of pupils that can move through the network per minute on this particular day. (2)

(Total 16 marks)

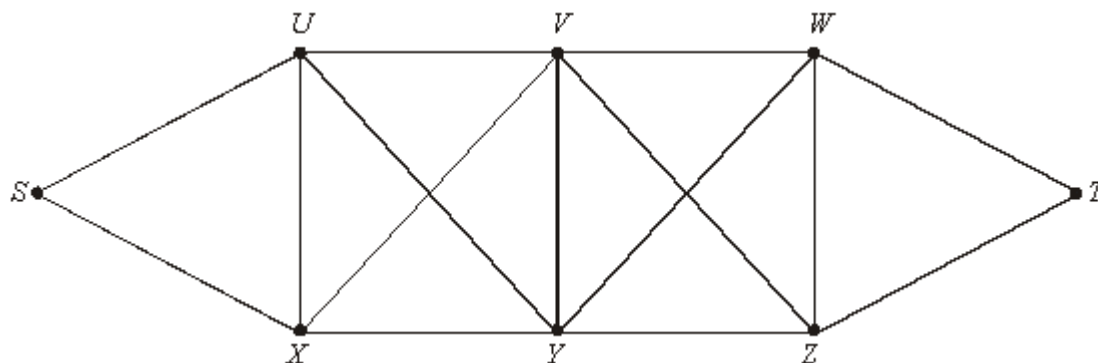
Q17.

The diagram shows a network of corridors in a building. People move steadily along the corridors from the entrance S towards the exit T . The weights on the edges show the maximum number of people per minute that can move along each corridor.



- (a) (i) Find the value of the cut marked C on the diagram. (1)
- (ii) Use your answer to part (a)(i) to make a deduction about the maximum flow of people through the building. (2)
- (b) State the maximum possible flows along the routes $SUYWT$ and $SXYZT$. (2)

- (c) (i) Taking your answers to part (b) as the initial flow, use a labelling procedure on **Figure 1** to find the maximum flow from S to T



Route	Flow
SUYWT	
SXYZT	

Figure 1

(6)

- (ii) State the value of the maximum flow and, on **Figure 2**, illustrate a possible flow along each edge corresponding to this maximum flow.

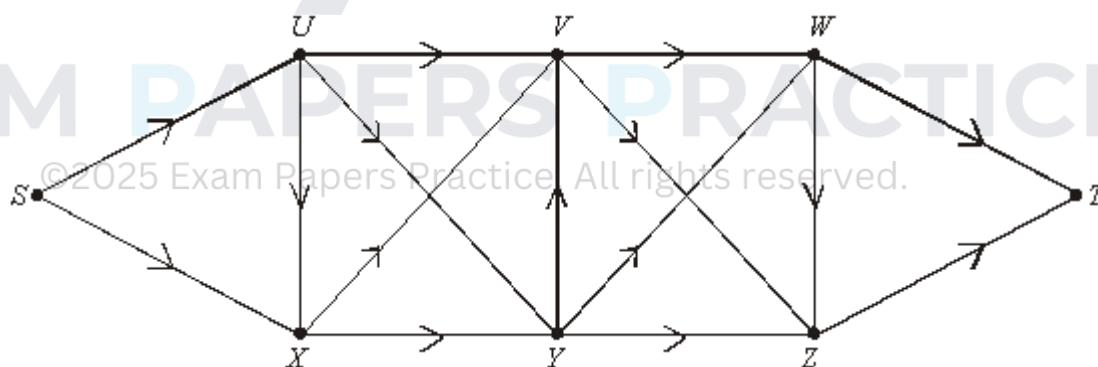


Figure 2

(2)

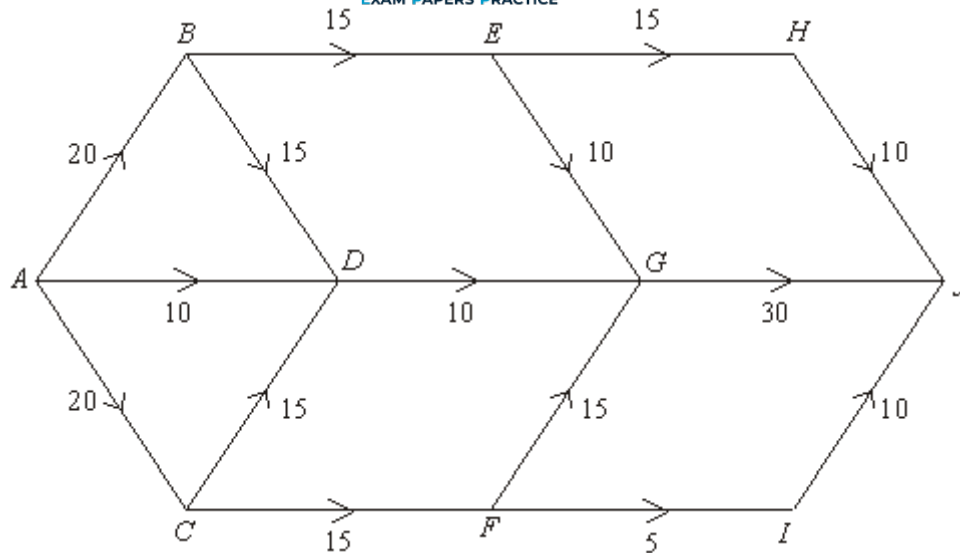
- (iii) Find a cut through the original network that has value equal to this maximum flow.

(2)

(Total 15 marks)

Q18.

The following diagram shows the maximum number of cars that can travel along each section of a one-way system in a 5 minute interval.



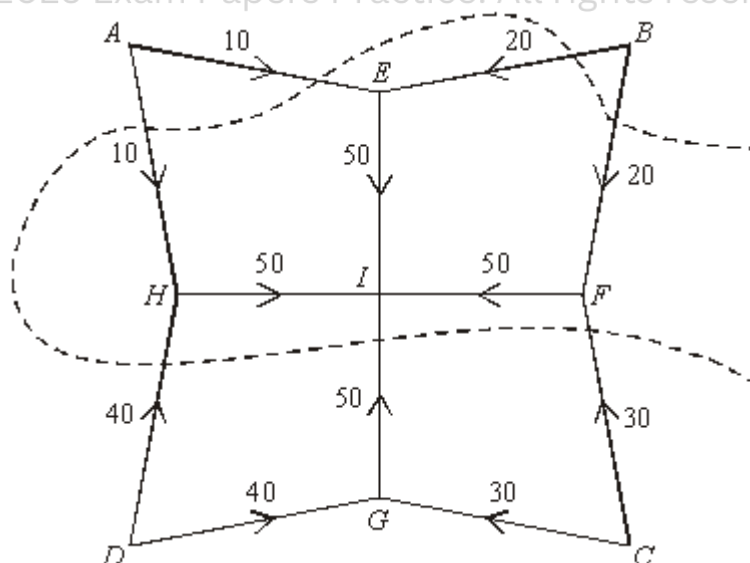
- (a) On the figure above, starting from a position of no flow of cars along the one-way system, use flow augmentation to find the maximum flow. (4)
- (b) Confirm that you have a maximum flow by finding a cut of the same value. (1)
- (c) A new road is to be added to the system. Determine which two vertices this new road should connect in order to make a flow of more than 50 cars, in a 5 minute interval, possible. (2)

(Total 7 marks)

Q19.

The following diagram represents a network of pipes used in a water system in a garden. The number on each arc represents the maximum volume of water, in cubic centimetres, that can flow along the corresponding pipe in one second.

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- (a) State the vertices that represent the sources. (1)

- (b) State the vertex that represents the sink. (1)
- (c) Find the value of the cut X . (2)
- (d) (i) On **Figure 1**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

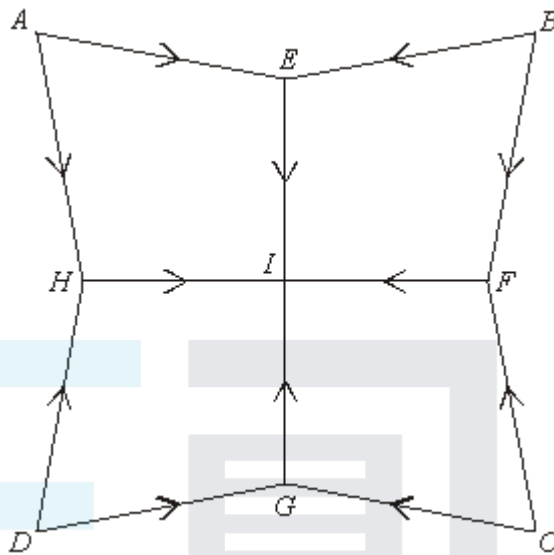


Figure 1

- (ii) State why this flow is maximum. (1)
- (e) On a particular day there is a restriction at vertex E which allows a maximum flow through E of 20.

On **Figure 2**, find, by inspection, the maximum flow through the network on this day.

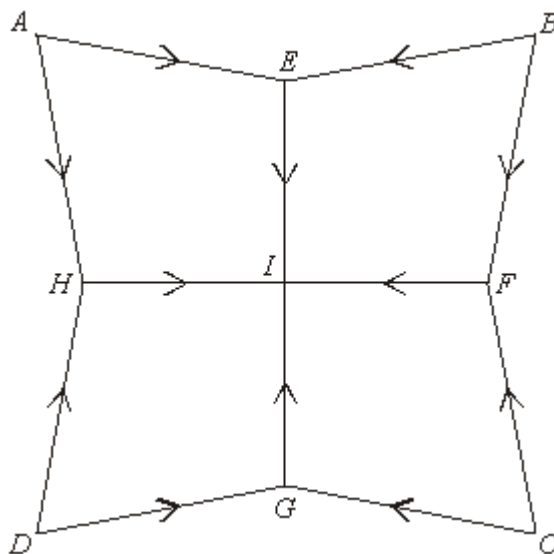
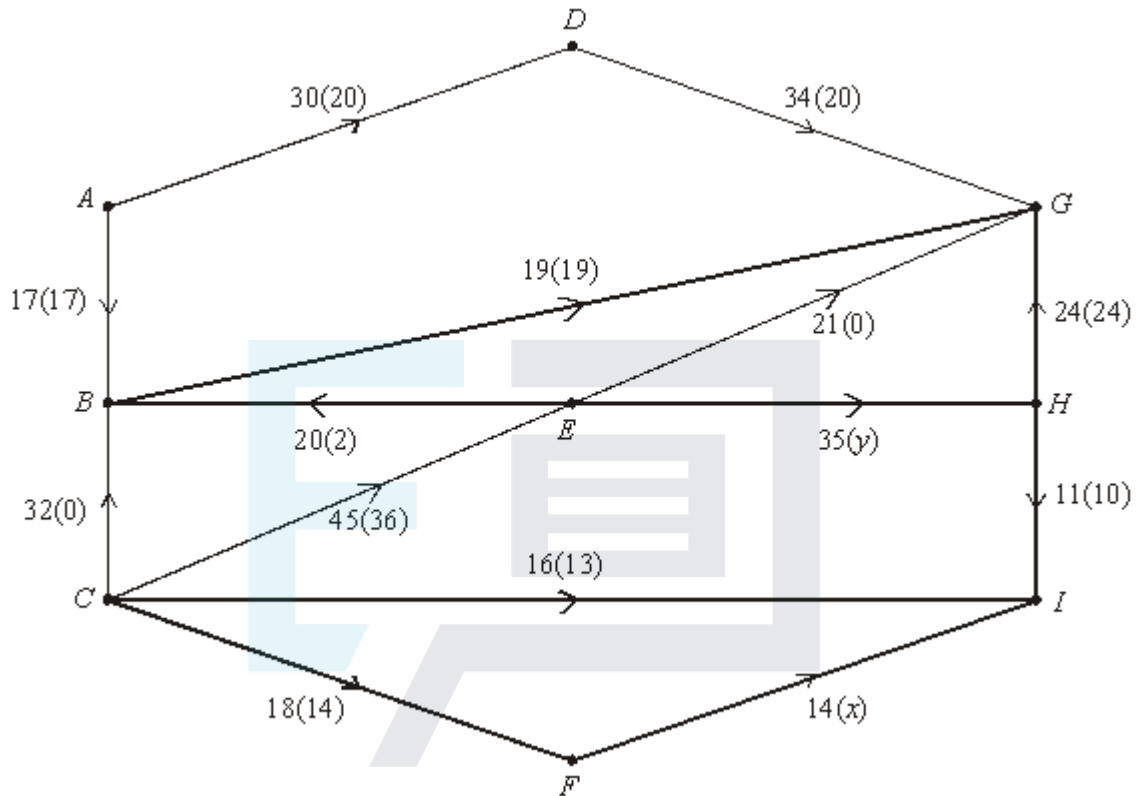


Figure 2

Q20.

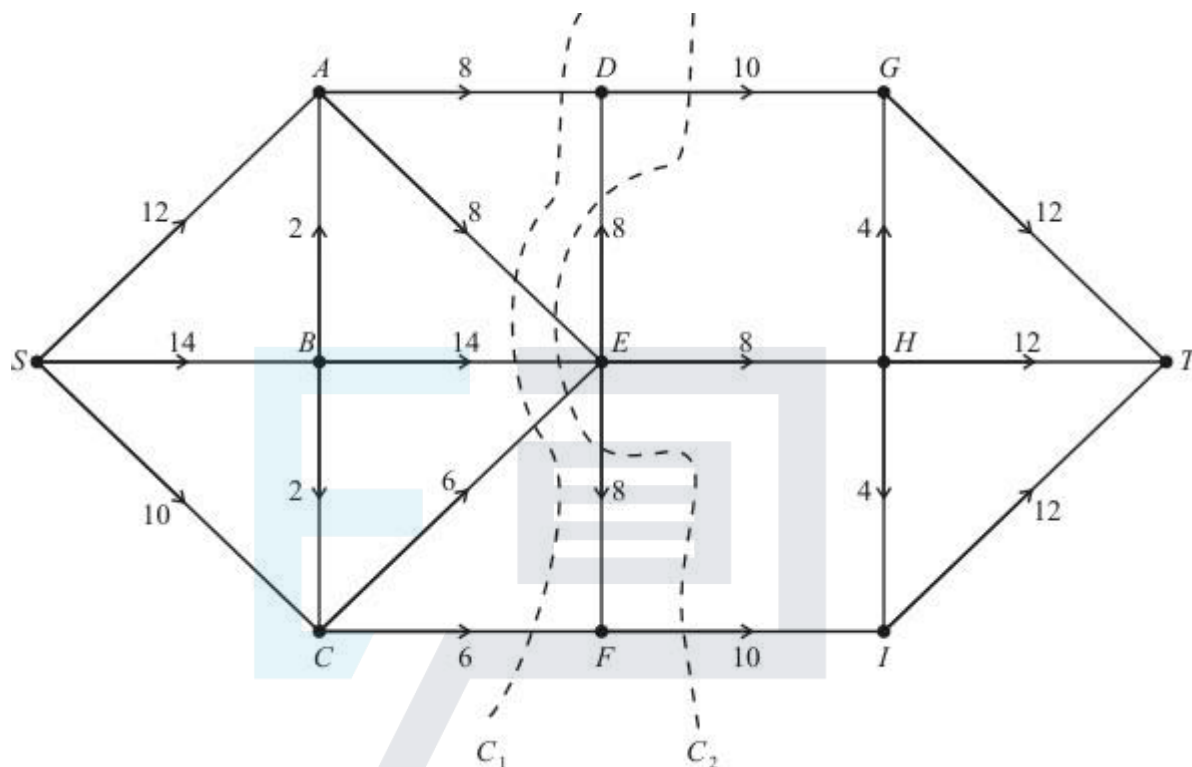
The following diagram shows a capacitated directed network. The number on each arc is the capacity of that arc. The numbers in brackets represent an initial feasible flow through the network.



- (a) List:
- (i) the source vertices; (1)
 - (ii) the sink vertices. (1)
- (b) Find the value of:
- (i) the initial flow; (1)
 - (ii) x ; (1)
 - (iii) y . (1)
- (c) On the diagram above, starting from the given initial flow, use flow augmentation to find a flow of 122. (6)

Q21.

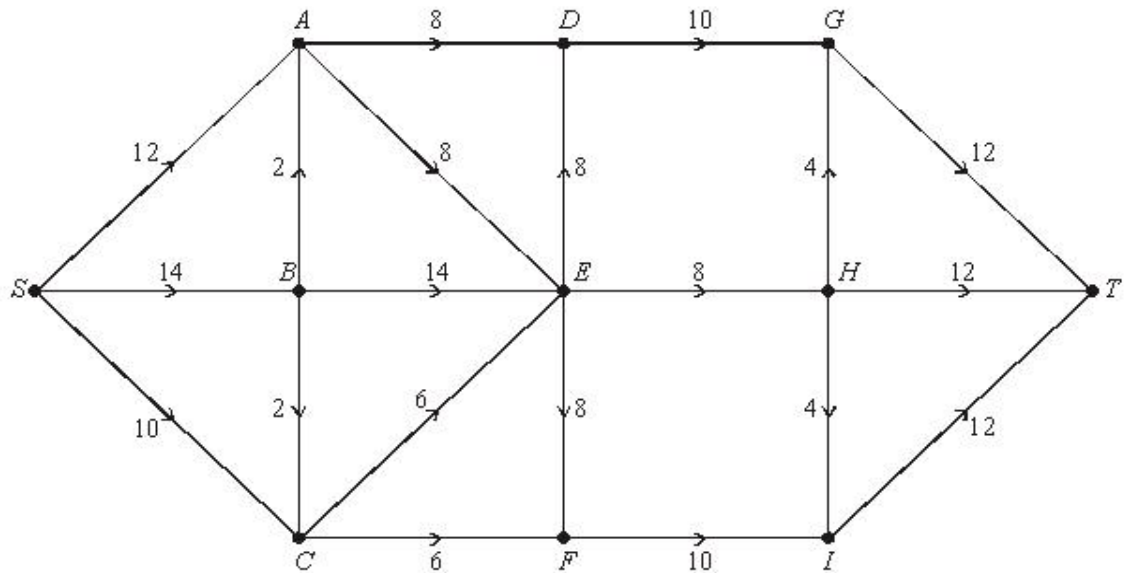
The following diagram shows a network of drainage pipes on a housing estate. Vertex S is the highest point on the estate and vertex T is the lowest point on the estate. The numbers on the edges show the maximum volume of water, in cubic metres, that can flow along each section of the pipe in one minute.



- (a) Find the value of the cuts marked C_1 and C_2 on the diagram.

(3)

- (b) On the diagram below, starting from a position of zero flow, use flow augmentation to find the maximum flow of water through the drainage pipes.



(7)
(Total 10 marks)

Q22.

A greengrocer has two suppliers, *A* and *B*, and three storage depots, *C*, *D* and *E*. He needs to transport his stock to three retail outlets *X*, *Y* and *Z*. The capacities of the possible routes, in van loads per week, are shown in the following diagram.



Figure 1

- (a) Add a super-source (*S*) and a super-sink (*W*) on **Figure 1** to obtain a single source,

single sink capacitated network. Show the capacities of each arc you have added.

(2)

- (b) State the maximum flow along the routes *SADYW* and *SBEZW*.

(2)

- (c)

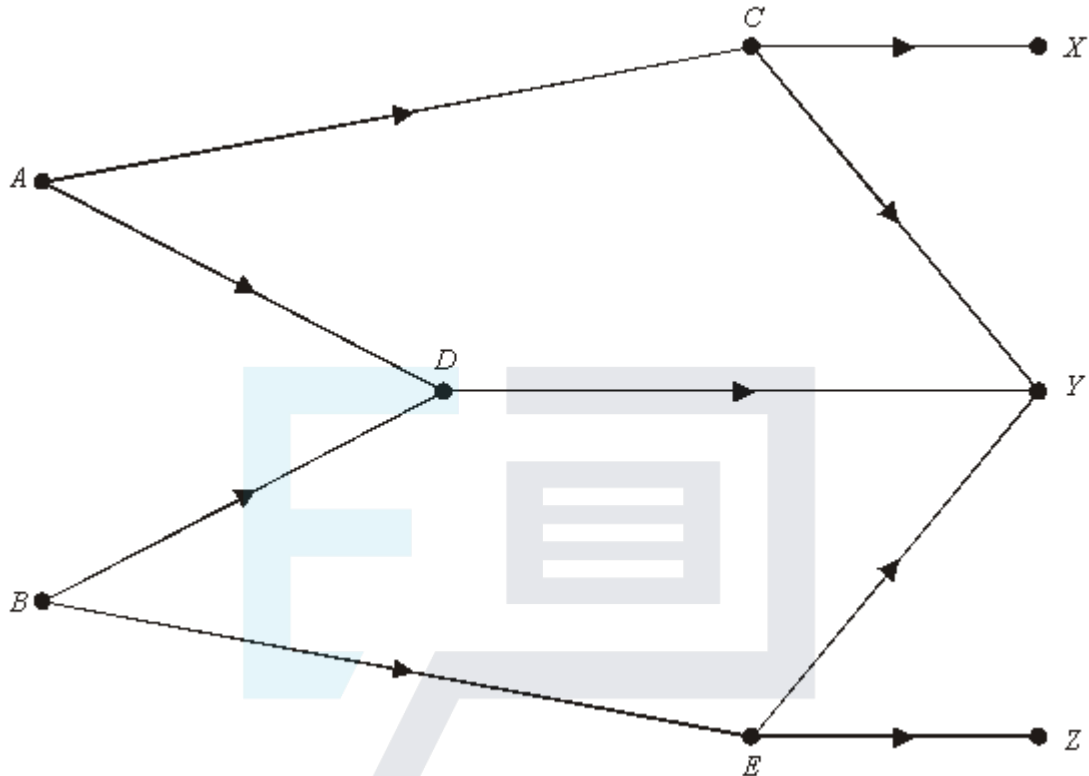


Figure 2

- (i) Show your answers to part (b) on **Figure 2** and, taking this as the initial flow pattern, use flow augmentation to find the maximum flow from *S* to *W*.

(6)

- (ii) Prove that your flow is maximal.

(2)

(Total 12 marks)

Q23.

- (a) When investigating a network flow problem, a student finds:

- (i) a cut with capacity 25 and a flow of 25;
- (ii) a cut with capacity 25 and a flow of 15.

In each case, write down what the student can deduce.

(2)

- (b) **Diagram 1**, below, shows a network flow problem. The numbers on the arcs represent the minimum and maximum flow along each arc respectively. By considering the flow at an appropriate vertex, explain why a flow is **not** possible

through this network.

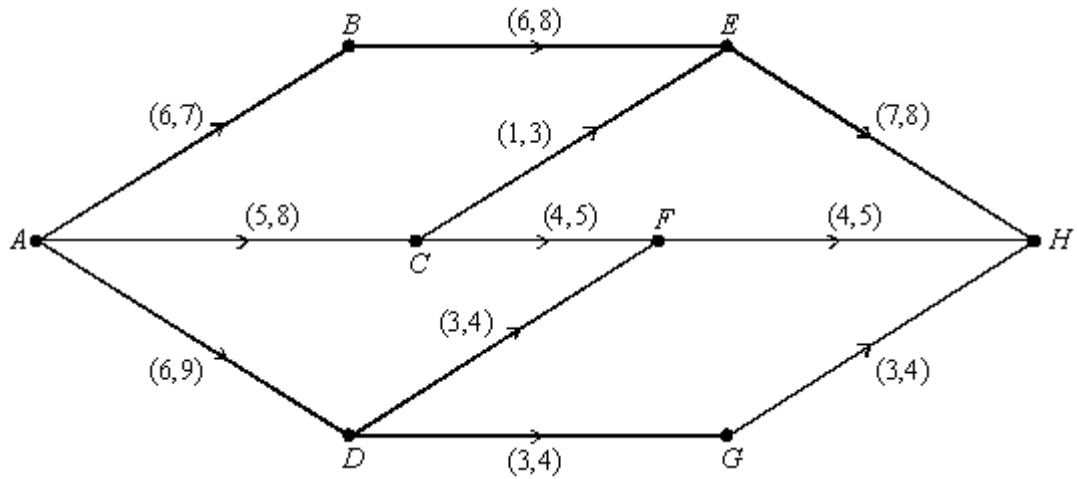


Diagram 1

(2)

- (c) **Diagram 2**, below, shows the maximum flow along each arc in a network flow problem.

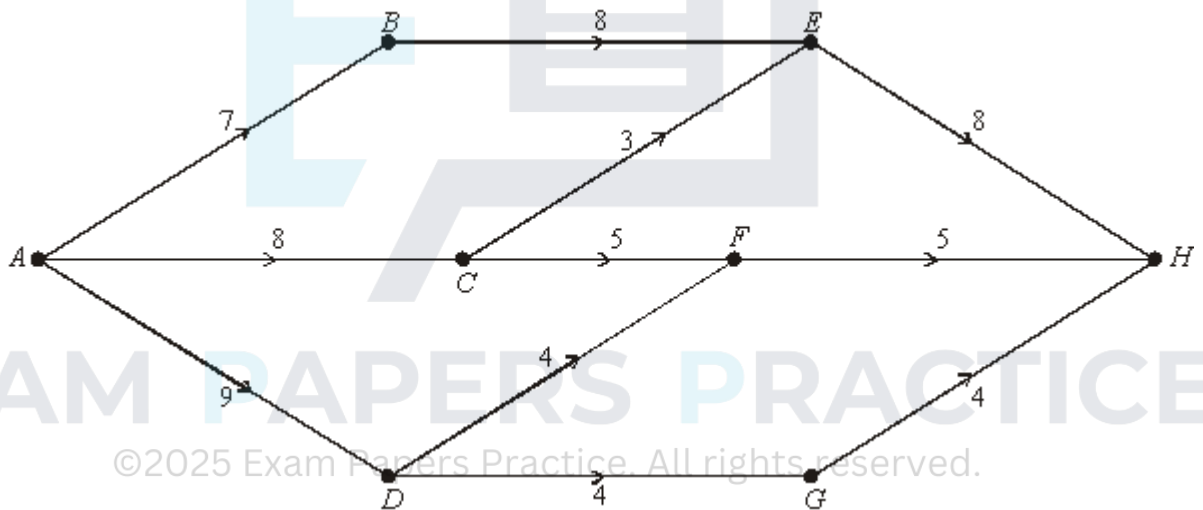


Diagram 2

- (i) On **Diagram 2**, starting from a position of zero flow, use flow augmentation to find the maximum flow.

(5)

- (ii) Confirm that you have a maximum flow by finding a cut of the same value.

(1)

(Total 10 marks)

Q24.

The following diagram shows the maximum number of cars that can travel along a one-way system in an hour.

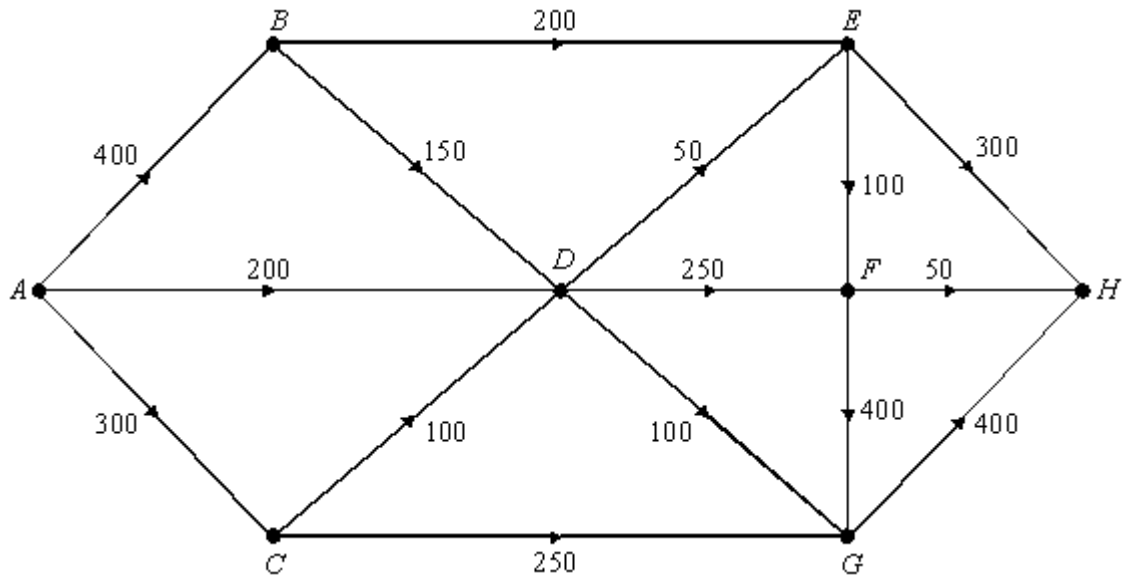


Figure 1

- (a) On **Figure 1**, starting from a position of no flow of cars along the one-way system, use flow augmentation to find the maximum flow. Show your final solution on **Figure 2**. (6)
- (b) Confirm that you have a maximum flow by finding a minimum cut of the same value. (2)

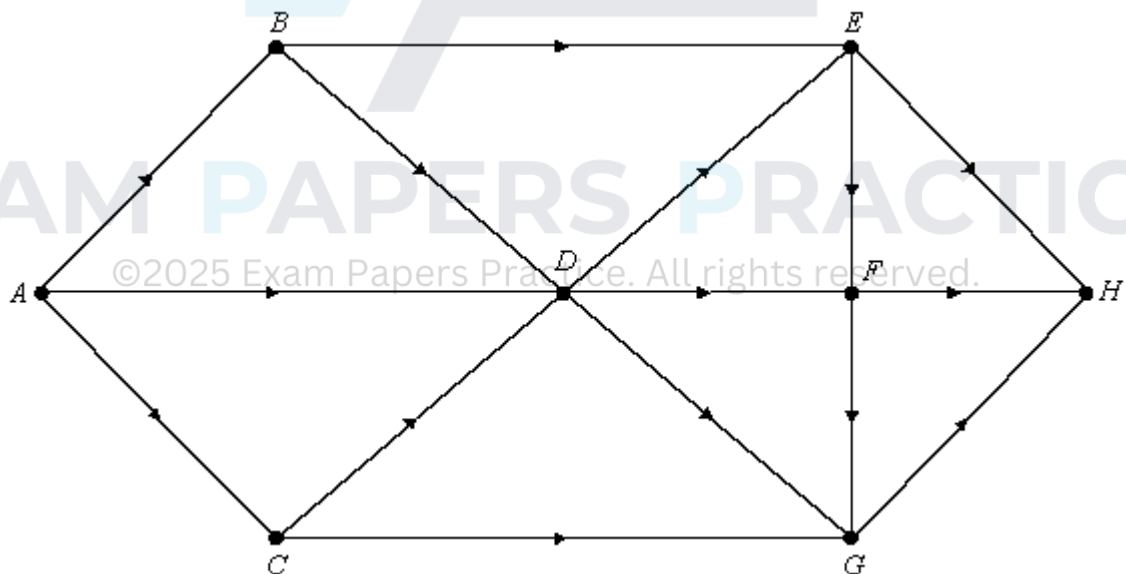
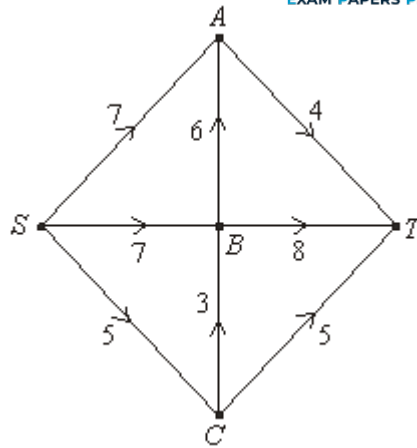


Figure 2

(Total 8 marks)

Q25.

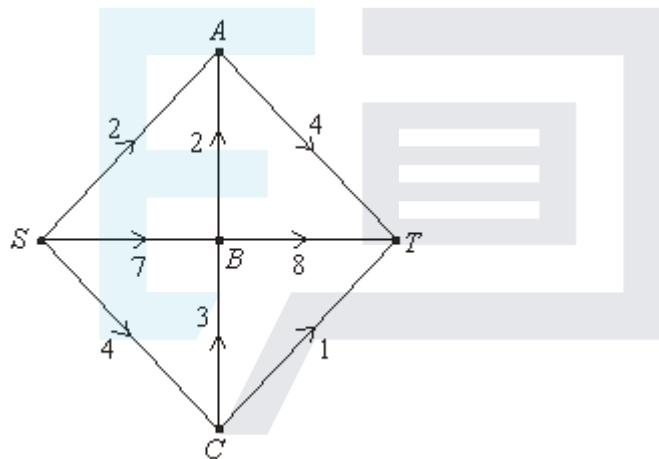
The diagram shows the capacities of the arcs of a network.



(a) Find a cut of 16 in the network above.

(2)

(b) The next diagram shows a flow of 13 from S to T in the same network.



(i) Find a flow-augmenting path from S to T which uses just one other vertex and which raises the given flow of 13 to a flow of 14.

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(2)

(ii) Find a further flow-augmenting path from S to T which raises the flow from 14 to 16.

(2)

(c) How do you know that the flow of 16 is the maximum possible in this network?

(1)

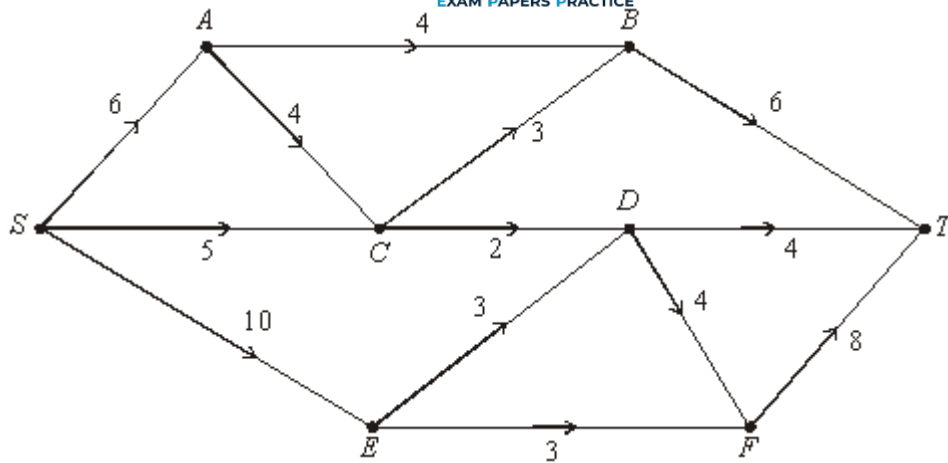
(d) The capacity of one arc of the network is to be increased in order to enable a flow of 19 from S to T . State which arc should be chosen and give a reason.

(3)

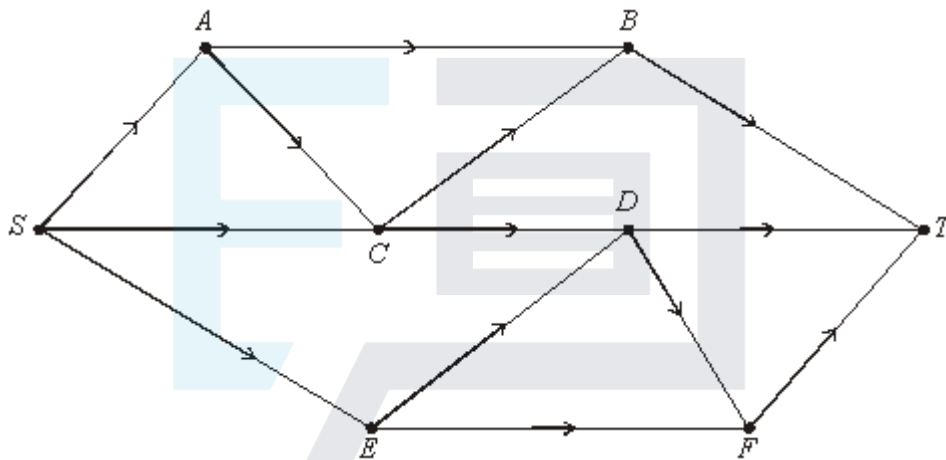
(Total 10 marks)

Q26.

The network shows a system of pipes and their capacities:



- (a) Starting from a zero flow, use flow-augmenting paths on the figure below to find a maximum flow from S to T .



(5)

- (b) Find a cut which confirms that your flow in (a) is a maximum.

(2)

- (c) (i) Show that in any flow from S to T there can be at most a flow of 6 in the pipe SE .

(1)

- (ii) Show that, if the pipe AC is saturated during a flow from S to T , there can be at most a flow of 1 in the pipe SC .

(2)

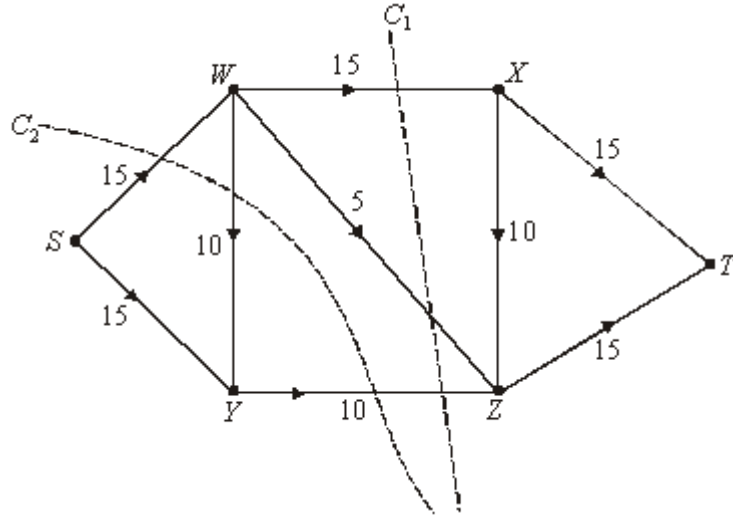
- (iii) Deduce that in any maximum flow from S to T the pipe AC will not be saturated.

(2)

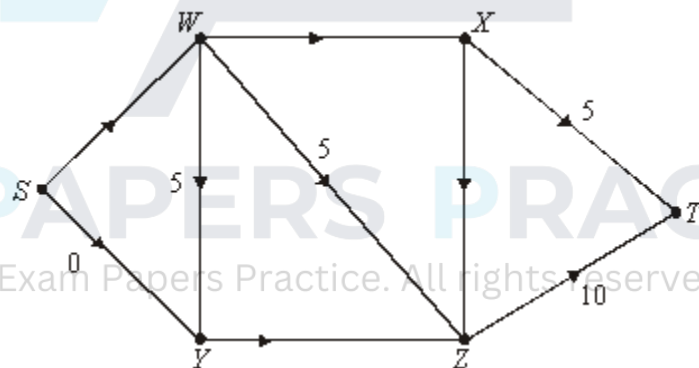
(Total 12 marks)

Q27.

In the network shown below, S is the source, T is the sink, and the number on each arc indicates its capacity:



- (a) Find the value of:
- (i) the cut C_1 ; (1)
 - (ii) the cut C_2 . (1)
- (b) The diagram below shows a flow from S to T in which the individual flows in four of the arcs have been omitted.

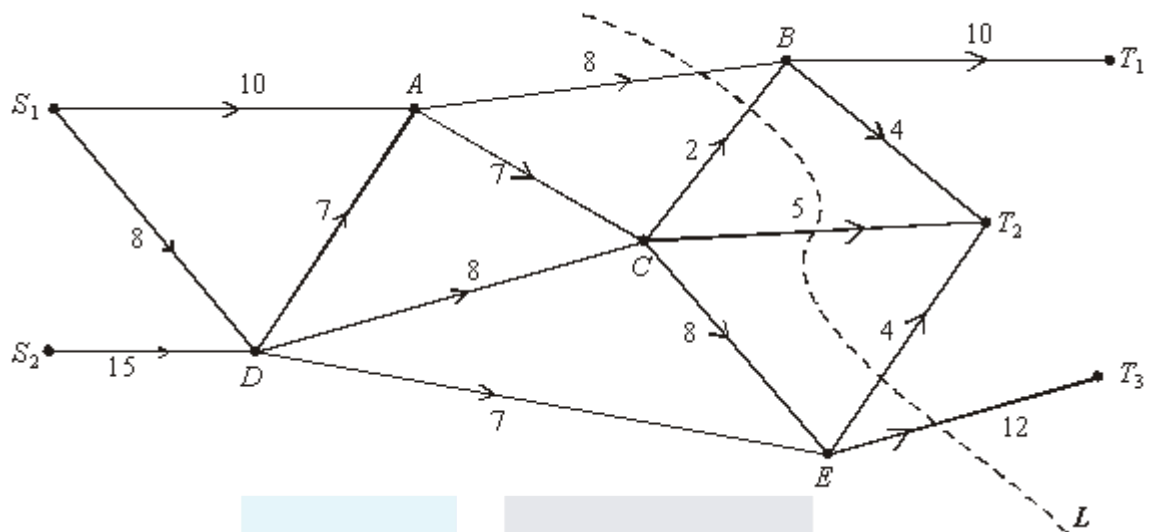


- Calculate the flows in YZ , XZ , WX and SW . (3)
- (c) Find two flow-augmenting paths, each of value 5, which together can be added to the flow illustrated in part (b). (4)
- (d) Explain how you know that no additional flow-augmenting paths exist. (2)
- (Total 11 marks)**

Q28.

Two food factories S_1 and S_2 supply three major supermarkets T_1 , T_2 and T_3 . The various routes from factories to supermarkets are shown in the network below, where the number

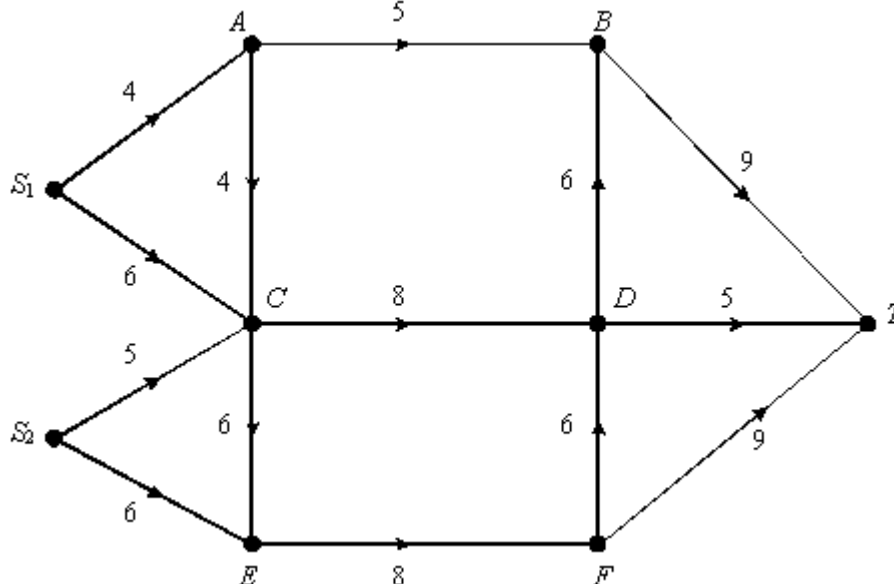
on each arc represents the daily maximum number of lorry-loads which can use that route.



- (a) Add a supersource S and supersink T to the diagram above, together with appropriate arcs and capacities, in order to create an equivalent network with a single source and single sink. (2)
 - (b) (i) Calculate the value of the cut shown by the broken line L above. (1)
 - (ii) Find a cut of lower value than the one given in part (i). (2)
 - (c) In your adapted network in the diagram above, start from a zero flow and use flow-augmenting paths to find a flow of value 30 from S to T . (6)
 - (d) State how you can be sure that the flow found in part (c) is a maximum flow. (2)
 - (e) Show that the three supermarkets cannot each be supplied with 10 lorry-loads in a day. (3)
- (Total 16 marks)**

Q29.

In the network illustrated, S_1 and S_2 are sources, T is the sink, and the number on each arc indicates its capacity.

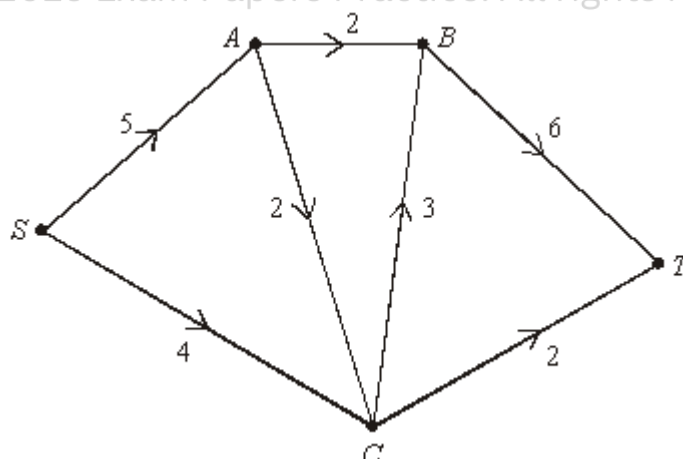


- On the diagram above add a super-source S and appropriate arcs and capacities in order to convert this network into one with a single source. (2)
 - By using flow augmentation, find a flow of 20 in this network. Mark the flow clearly on the diagram above. (5)
 - By finding an appropriate cut, explain how you know that a flow of 20 is the maximum possible. (3)
- (Total 10 marks)

Q30.

In the network below the number on each arc represents the maximum possible flow in that arc.

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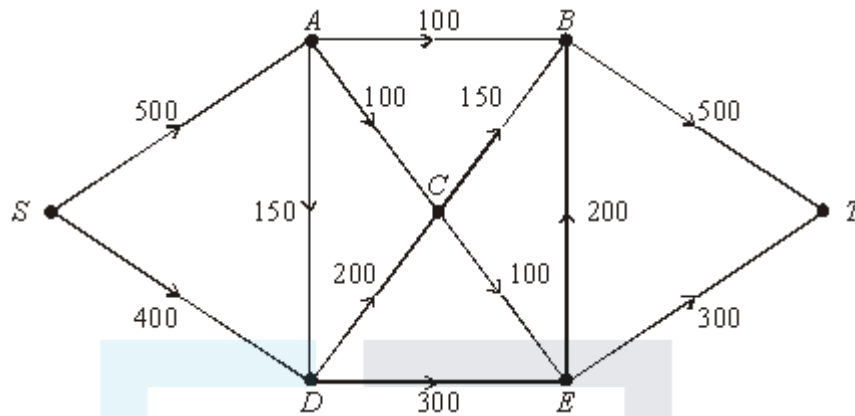
- Starting from a zero flow in this network, use flow-augmenting paths to find a maximum flow from S to T . (4)
- Find a cut in the network which proves that the flow which you found in part (a)(i) is

a maximum flow.

(2)
(Total 6 marks)

Q31.

The following network shows a one-way system of roads: the number on each arc is the maximum number of cars which can travel along that road in one hour.



- (a) Starting from a zero flow, carry out repeated augmentations until you find a maximum flow from S to T . Show your workings on **Figure 1** and give full details of the maximum flow on **Figure 2**.

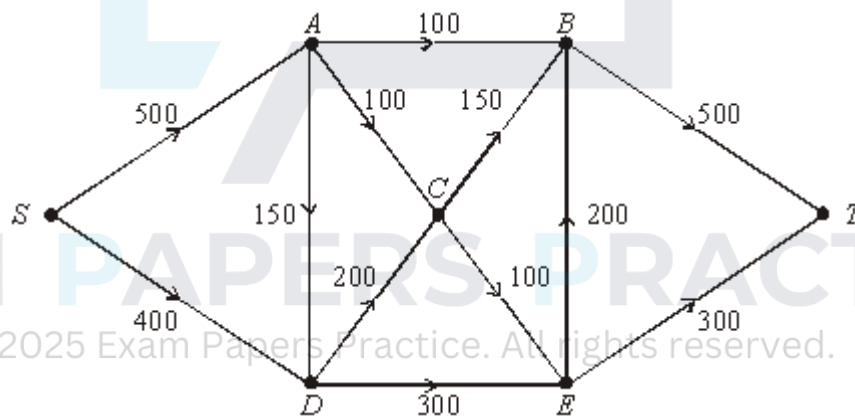


Figure 1

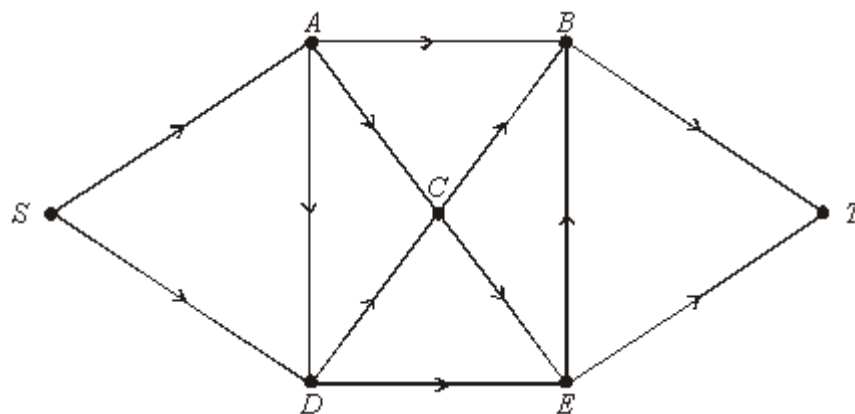


Figure 2

(6)

- (b) Confirm that the flow which you found in part (a) is a maximum by finding a cut of the same value. (2)
- (c) A large underpass is to be built to take traffic directly from C to T . In the corresponding new network state how to augment your answer to part (a) in order to increase the flow by 50 cars per hour. Show, however, that no further increase is possible. (3)

(Total 11 marks)



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